

# Guana River Marsh Aquatic Preserve

## SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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## Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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## Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR\_QAQCFlagCode* and *SEACAR\_QAQC\_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

| <i>Parameter Name</i>                     | <i>Units</i> | <i>Low Threshold</i> | <i>High Threshold</i> |
|-------------------------------------------|--------------|----------------------|-----------------------|
| Chlorophyll a, Uncorrected for Pheophytin | ug/L         | -                    | -                     |
| Dissolved Oxygen                          | mg/L         | -0.000001            | 50                    |
| Dissolved Oxygen Saturation               | %            | -0.000001            | 500                   |
| Fluorescent Dissolved Organic Matter      | QSE          | -                    | -                     |
| Salinity                                  | ppt          | -0.000001            | 70                    |
| Specific Conductivity                     | mS/cm        | -0.000001            | 200                   |
| Turbidity                                 | NTU          | -0.000001            | 4000                  |
| Water Temperature                         | Degrees C    | -5.000000            | 45                    |
| pH                                        | None         | 2.000000             | 14                    |

Table 2: Discrete Water Quality threshold values

| <i>Parameter Name</i>     | <i>Units</i> | <i>Low Threshold</i> | <i>High Threshold</i> |
|---------------------------|--------------|----------------------|-----------------------|
| Ammonia, Un-ionized (NH3) | mg/L         | -                    | -                     |
| Ammonium (NH4)            | mg/L         | -                    | -                     |

| <i>Parameter Name</i>                     | <i>Units</i>    | <i>Low Threshold</i> | <i>High Threshold</i> |
|-------------------------------------------|-----------------|----------------------|-----------------------|
| Chlorophyll a, Corrected for Pheophytin   | ug/L            | -                    | -                     |
| Chlorophyll a, Uncorrected for Pheophytin | ug/L            | -                    | -                     |
| Colored Dissolved Organic Matter          | PCU             | -                    | -                     |
| Dissolved Oxygen                          | mg/L            | -0.000001            | 25                    |
| Dissolved Oxygen Saturation               | %               | -0.000001            | 310                   |
| Fluorescent Dissolved Organic Matter      | QSE             | -                    | -                     |
| Light Extinction Coefficient              | m <sup>-1</sup> | -                    | -                     |
| NO2+3, Filtered                           | mg/L            | -                    | -                     |
| Nitrate (NO3)                             | mg/L            | -                    | -                     |
| Nitrite (NO2)                             | mg/L            | -                    | -                     |
| Nitrogen, inorganic                       | mg/L            | -                    | -                     |
| Nitrogen, organic                         | mg/L            | -                    | -                     |
| Phosphate, Filtered (PO4)                 | mg/L            | -                    | -                     |
| Salinity                                  | ppt             | -0.000001            | 70                    |
| Secchi Depth                              | m               | 0.000001             | 50                    |
| Specific Conductivity                     | mS/cm           | 0.005000             | 100                   |
| Total Ammonia (N)                         | mg/L            | -                    | -                     |
| Total Kjeldahl Nitrogen                   | mg/L            | -                    | -                     |
| Total Nitrogen                            | mg/L            | -                    | -                     |
| Total Nitrogen                            | mg/L            | -                    | -                     |
| Total Phosphorus                          | mg/L            | -                    | -                     |
| Total Suspended Solids                    | mg/L            | -                    | -                     |
| Turbidity                                 | NTU             | -                    | -                     |
| Water Temperature                         | Degrees C       | 3.000000             | 40                    |
| pH                                        | None            | 2.000000             | 13                    |

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

| <i>SEACAR QAQC Description</i>           | <i>Include</i> | <i>SEACAR QAQCFlagCode</i> |
|------------------------------------------|----------------|----------------------------|
| Exceeds maximum threshold                | 0              | 2Q                         |
| Below minimum threshold                  | 0              | 4Q                         |
| Within threshold tolerance               | 1              | 6Q                         |
| No defined thresholds for this parameter | 1              | 7Q                         |

## Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

### STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

| <i>Qualifier Source</i> | <i>Value Qualifier</i> | <i>Include</i> | <i>MDL</i> | <i>Description</i>                                                       |
|-------------------------|------------------------|----------------|------------|--------------------------------------------------------------------------|
| STORET-WIN              | H                      | 0              | 0          | Value based on field kit determination; results may not be accurate      |
| STORET-WIN              | J                      | 0              | 0          | Estimated value                                                          |
| STORET-WIN              | V                      | 0              | 0          | Analyte was detected at or above method detection limit                  |
| STORET-WIN              | Y                      | 0              | 0          | Lab analysis from an improperly preserved sample; data may be inaccurate |

## Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476 - Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network** and data flagged with Value Qualifier **H** are included for this program only.

**H** - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

**I** - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

**Q** - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

**S** - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

**U** - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

## Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

| <i>Qualifier Source</i> | <i>Value Qualifier</i> | <i>Include</i> | <i>Description</i>                                                                           |
|-------------------------|------------------------|----------------|----------------------------------------------------------------------------------------------|
| SWMP                    | -1                     | 1              | Optional parameter not collected                                                             |
| SWMP                    | -2                     | 0              | Missing data                                                                                 |
| SWMP                    | -3                     | 0              | Data rejected due to QA/QC                                                                   |
| SWMP                    | -4                     | 0              | Outside low sensor range                                                                     |
| SWMP                    | -5                     | 0              | Outside high sensor range                                                                    |
| SWMP                    | 0                      | 1              | Passed initial QA/QC checks                                                                  |
| SWMP                    | 1                      | 0              | Suspect data                                                                                 |
| SWMP                    | 2                      | 1              | Reserved for future use                                                                      |
| SWMP                    | 3                      | 1              | Calculated data: non-vented depth/level sensor correction for changes in barometric pressure |
| SWMP                    | 4                      | 1              | Historical: Pre-auto QA/QC                                                                   |
| SWMP                    | 5                      | 1              | Corrected data                                                                               |

## Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

## Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use\_In\_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use\_In\_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when  $p < 0.05$ ), or "Non-significant Trend" (when  $p \geq 0.05$ ). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

## Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined\_WQ\_WC\_NUT\_Chlorophyll\_a\_corrected\_for\_pheophytin-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Chlorophyll\_a\_uncorrected\_for\_pheophytin-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Colored\_dissolved\_organic\_matter\_CDOM-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Dissolved\_Oxygen-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Dissolved\_Oxygen\_Saturation-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_pH-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Salinity-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Secchi\_Depth-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Total\_Nitrogen-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Total\_Phosphorus-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Total\_Suspended\_Solids\_TSS-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Turbidity-2026-May-18.txt*
- *Combined\_WQ\_WC\_NUT\_Water\_Temperature-2026-May-18.txt*

# Chlorophyll a, Corrected for Pheophytin - Discrete

## Seasonal Kendall-Tau Trend Analysis

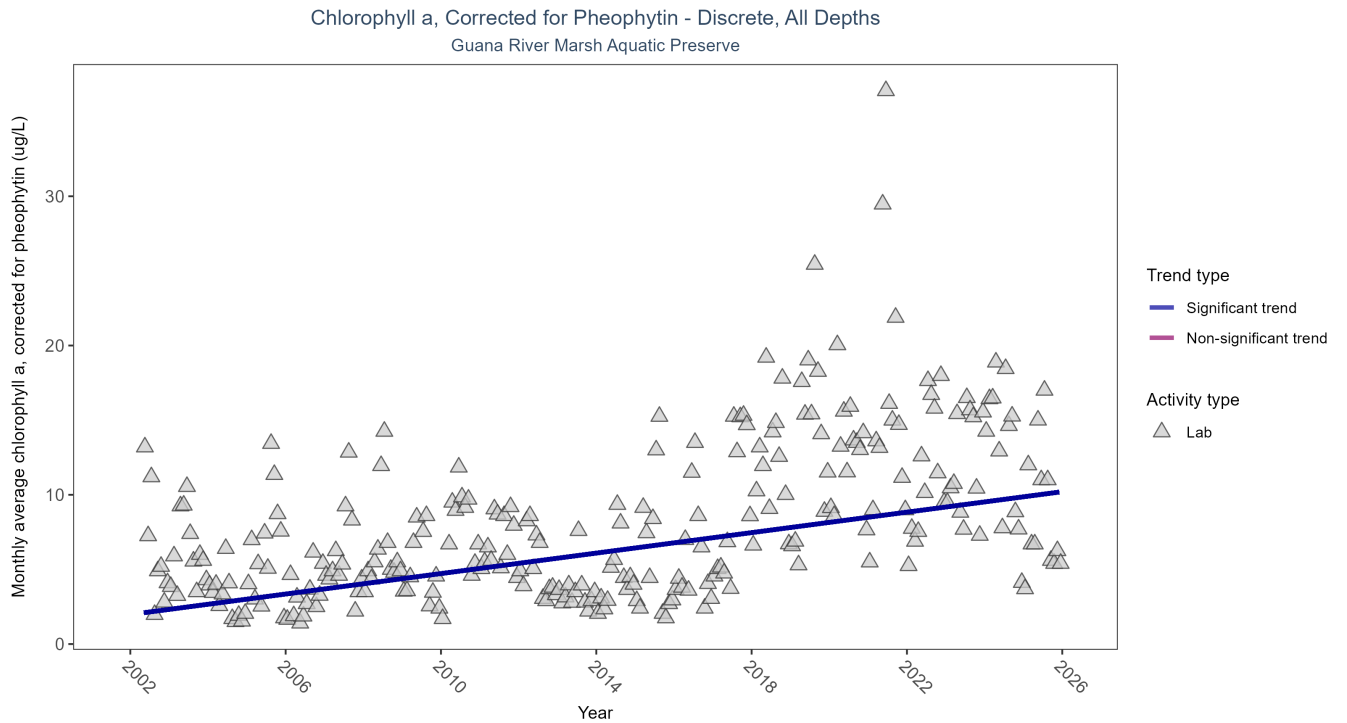


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|--------|---------------|-----------|---|
| Lab           | Increasing trend  | 1713           | 24                  | 2002 - 2025      | 5.5                 | 0.4174 | 1.979         | 0.3429    | 0 |

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.34  $\mu\text{g/L}$  per year, indicating a decrease in water clarity.



Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 4054      | 732    | 2002    | 2025    |
| 5014      | 630    | 2017    | 2024    |
| 5002      | 377    | 2002    | 2025    |

#### Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

5002 - Florida STORET / WIN<sup>3</sup>

#### Chlorophyll a, Uncorrected for Pheophytin - Discrete

#### Seasonal Kendall-Tau Trend Analysis

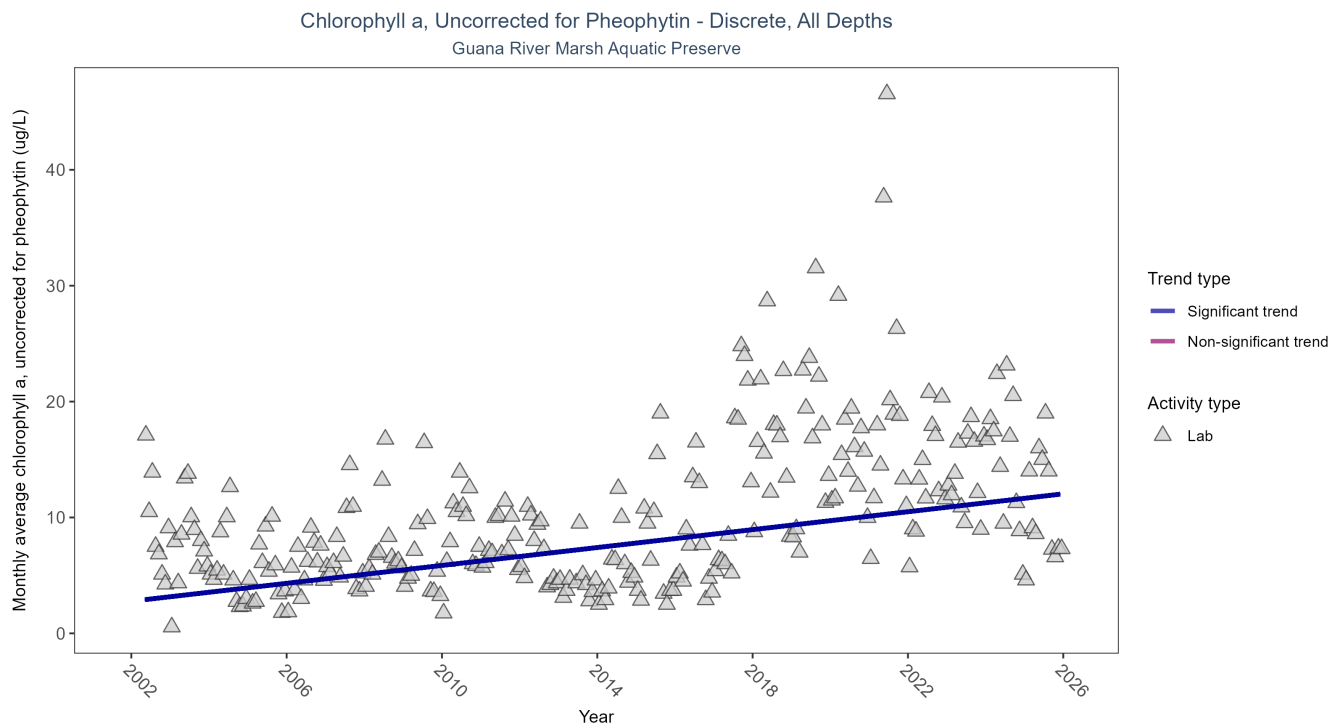


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|--------|---------------|-----------|---|
| Lab           | Increasing trend  | 1352           | 24                  | 2002 - 2025      | 7.8                 | 0.3988 | 2.7733        | 0.386     | 0 |

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.39 µg/L per year, indicating a decrease in water clarity.

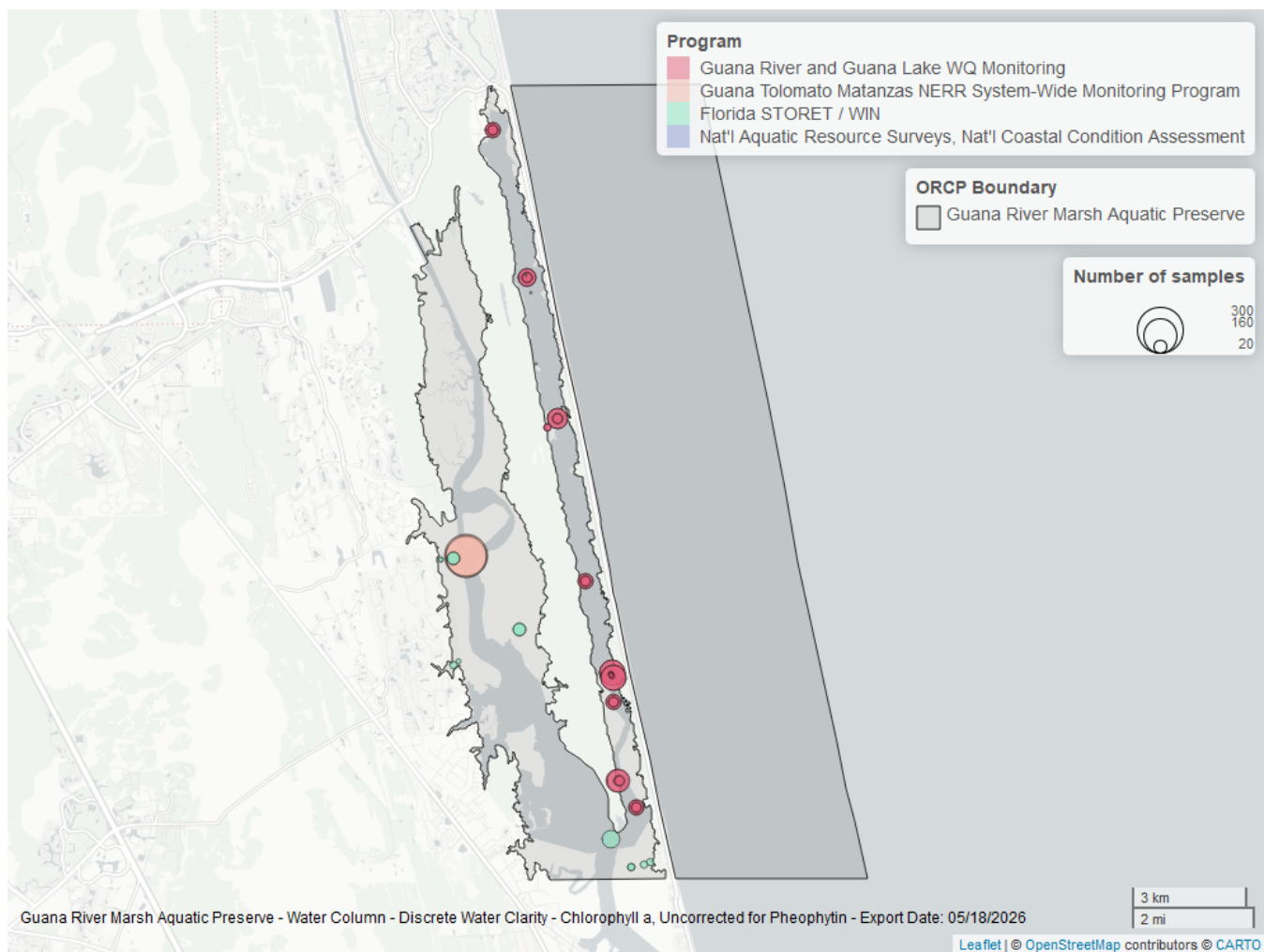


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

| ProgramID            | N_Data | YearMin | YearMax |
|----------------------|--------|---------|---------|
| <a href="#">5014</a> | 677    | 2017    | 2024    |
| <a href="#">4054</a> | 554    | 2002    | 2025    |
| 5002                 | 143    | 2019    | 2025    |
| 118                  | 1      | 2006    | 2006    |

#### Program names:

[5014](#) - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

[4054](#) - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5002 - Florida STORET / WIN<sup>3</sup>

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment<sup>4</sup>

#### Colored Dissolved Organic Matter - Discrete

#### Seasonal Kendall-Tau Trend Analysis

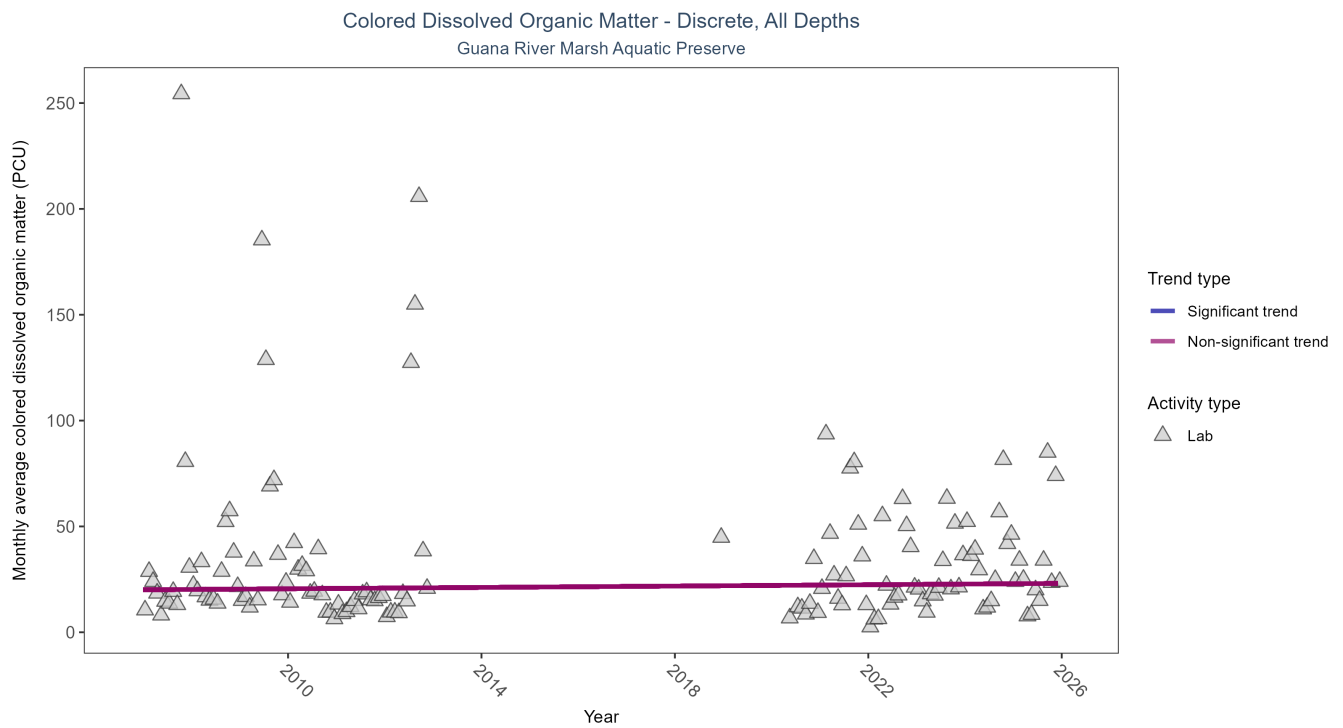


Figure 5: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Colored Dissolved Organic Matter

| Activity Type | Statistical Trend   | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P      |
|---------------|---------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|--------|
| Lab           | No detectable trend | 271            | 13                  | 2007 - 2025      | 20                  | 0.054 | 20.0715       | 0.1588    | 0.3939 |

Colored dissolved organic matter showed no detectable trend between 2007 and 2025.

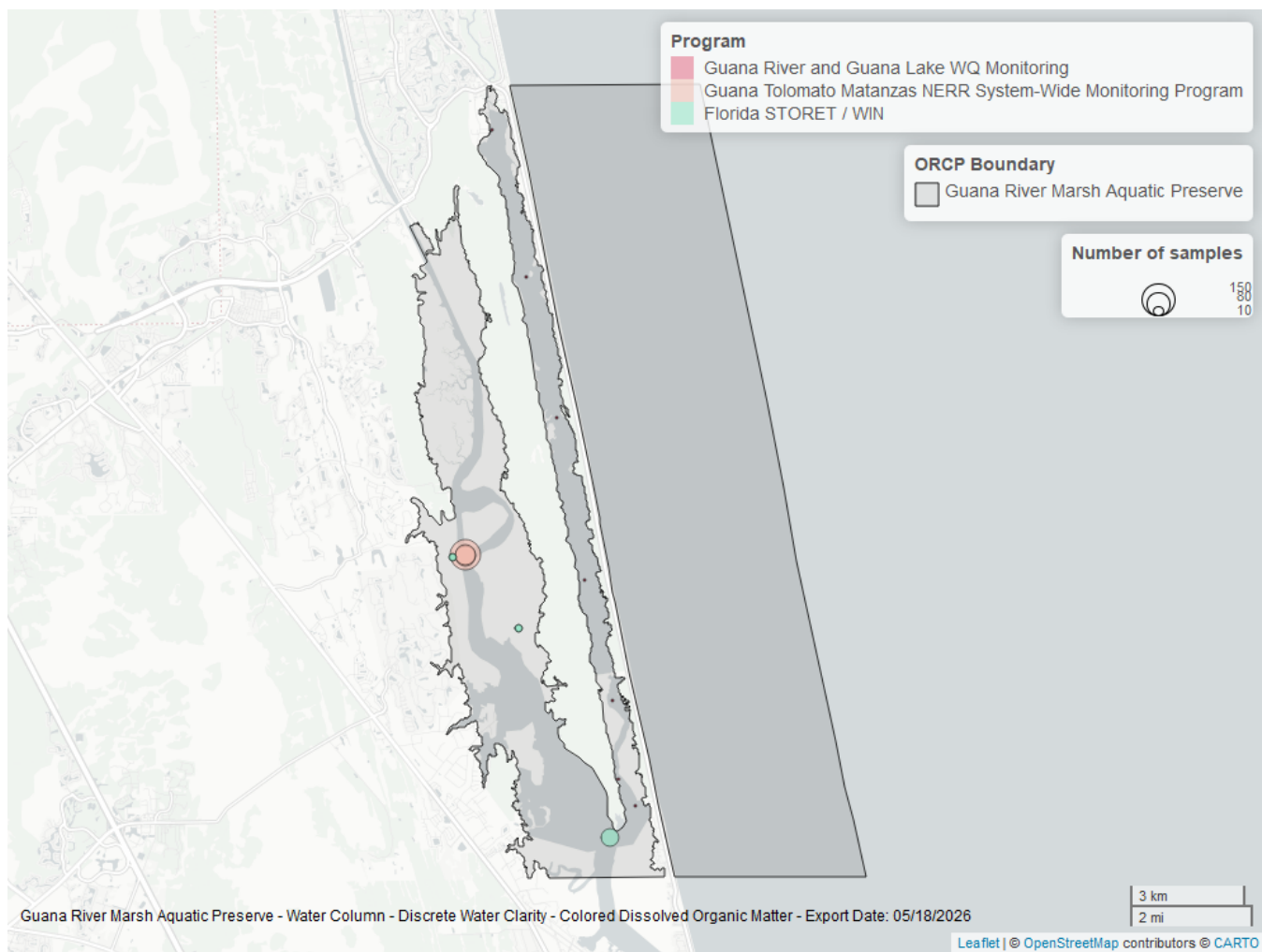


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Colored Dissolved Organic Matter

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 4054      | 200    | 2007    | 2025    |
| 5002      | 66     | 2020    | 2025    |
| 5014      | 7      | 2018    | 2018    |

#### Program names:

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5002 - Florida STORET / WIN<sup>3</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

#### Dissolved Oxygen - Discrete

#### Seasonal Kendall-Tau Trend Analysis

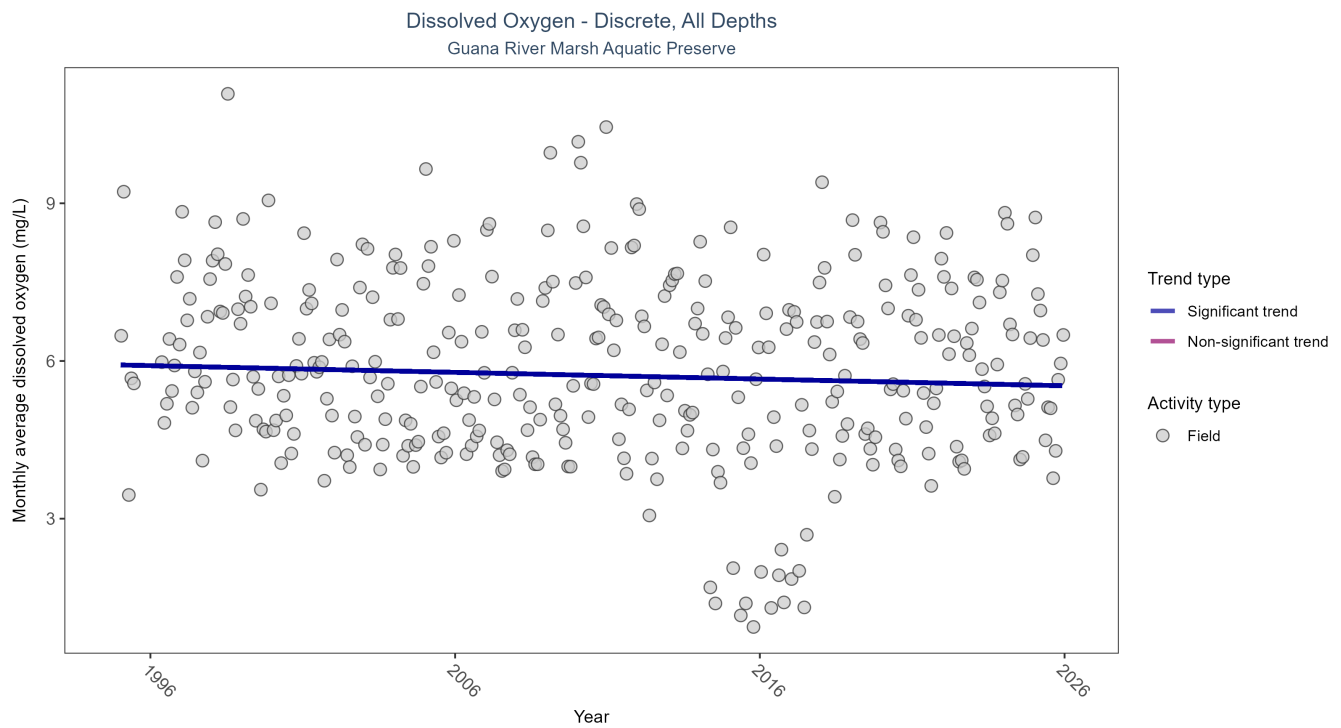


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|---------|---------------|-----------|--------|
| Field         | Decreasing trend  | 7686           | 31                  | 1995 - 2025      | 5.7                 | -0.0798 | 5.9243        | -0.0128   | 0.0314 |

Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

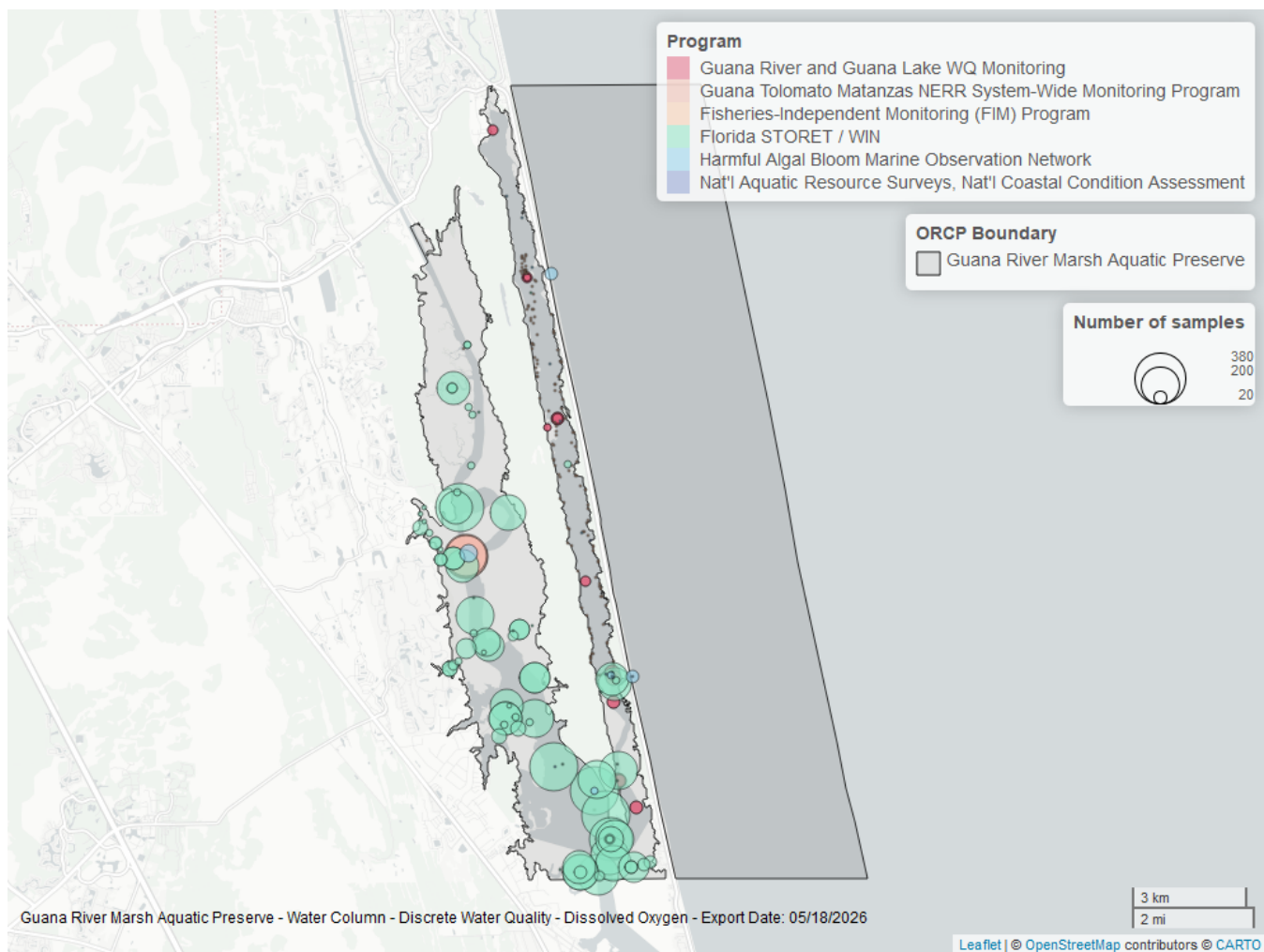


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for Dissolved Oxygen

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 6662   | 1995    | 2025    |
| 4054      | 534    | 2002    | 2025    |
| 5014      | 295    | 2017    | 2024    |
| 69        | 149    | 2001    | 2010    |
| 95        | 130    | 2007    | 2018    |
| 118       | 1      | 2006    | 2006    |

**Program names:**

5002 - Florida STORET / WIN<sup>3</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

69 - Fisheries-Independent Monitoring (FIM) Program<sup>5</sup>

95 - Harmful Algal Bloom Marine Observation Network<sup>6</sup>

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment<sup>4</sup>

## Dissolved Oxygen Saturation - Discrete

### Seasonal Kendall-Tau Trend Analysis

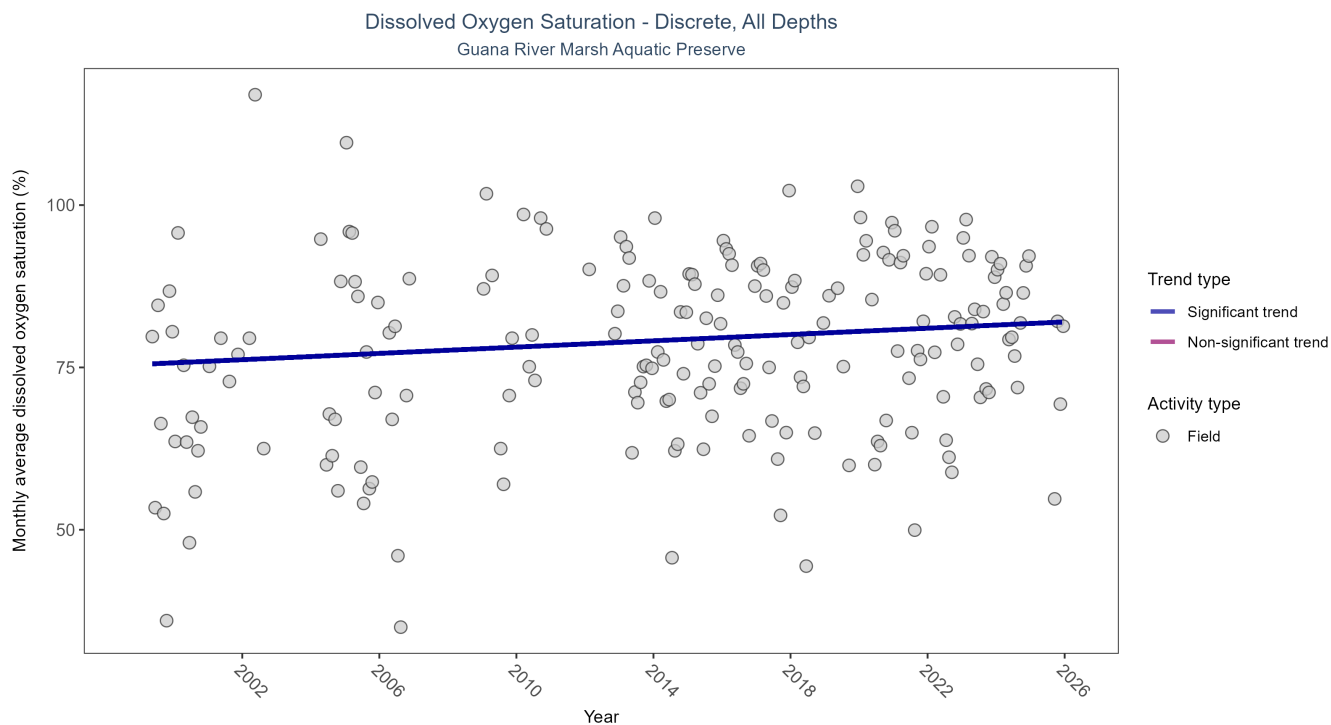


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen Saturation

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|--------|---------------|-----------|--------|
| Field         | Increasing trend  | 1215           | 23                  | 1999 - 2025      | 78.9                | 0.1159 | 75.4625       | 0.2424    | 0.0212 |

Monthly average dissolved oxygen saturation increased by 0.24% per year.



Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Dissolved Oxygen Saturation

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 1010   | 1999    | 2025    |
| 5014      | 220    | 2017    | 2022    |
| 95        | 3      | 2012    | 2013    |

#### Program names:

5002 - Florida STORET / WIN<sup>3</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

95 - Harmful Algal Bloom Marine Observation Network<sup>6</sup>

#### pH - Discrete

#### Seasonal Kendall-Tau Trend Analysis

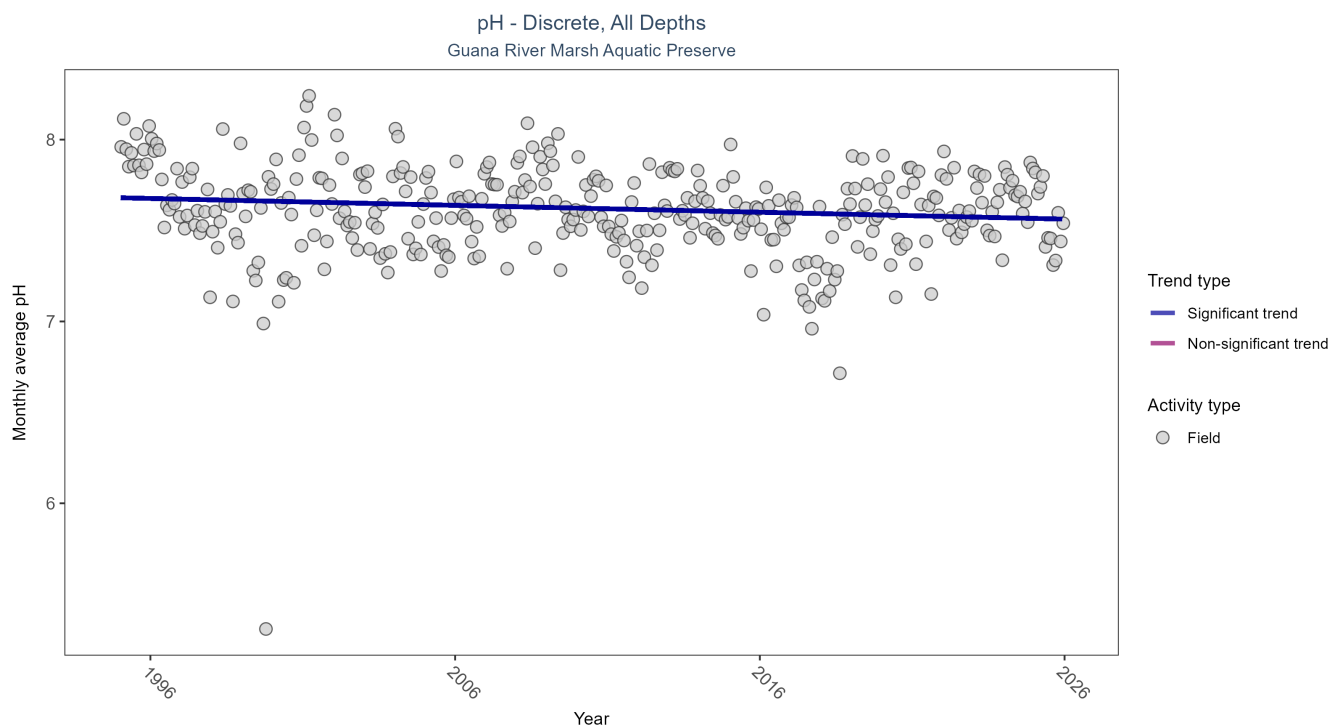


Figure 11: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for pH

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|---------|---------------|-----------|--------|
| Field         | Decreasing trend  | 6261           | 31                  | 1995 - 2025      | 7.7                 | -0.1217 | 7.6802        | -0.0038   | 0.0009 |

Monthly average pH decreased by less than 0.01 pH units per year.

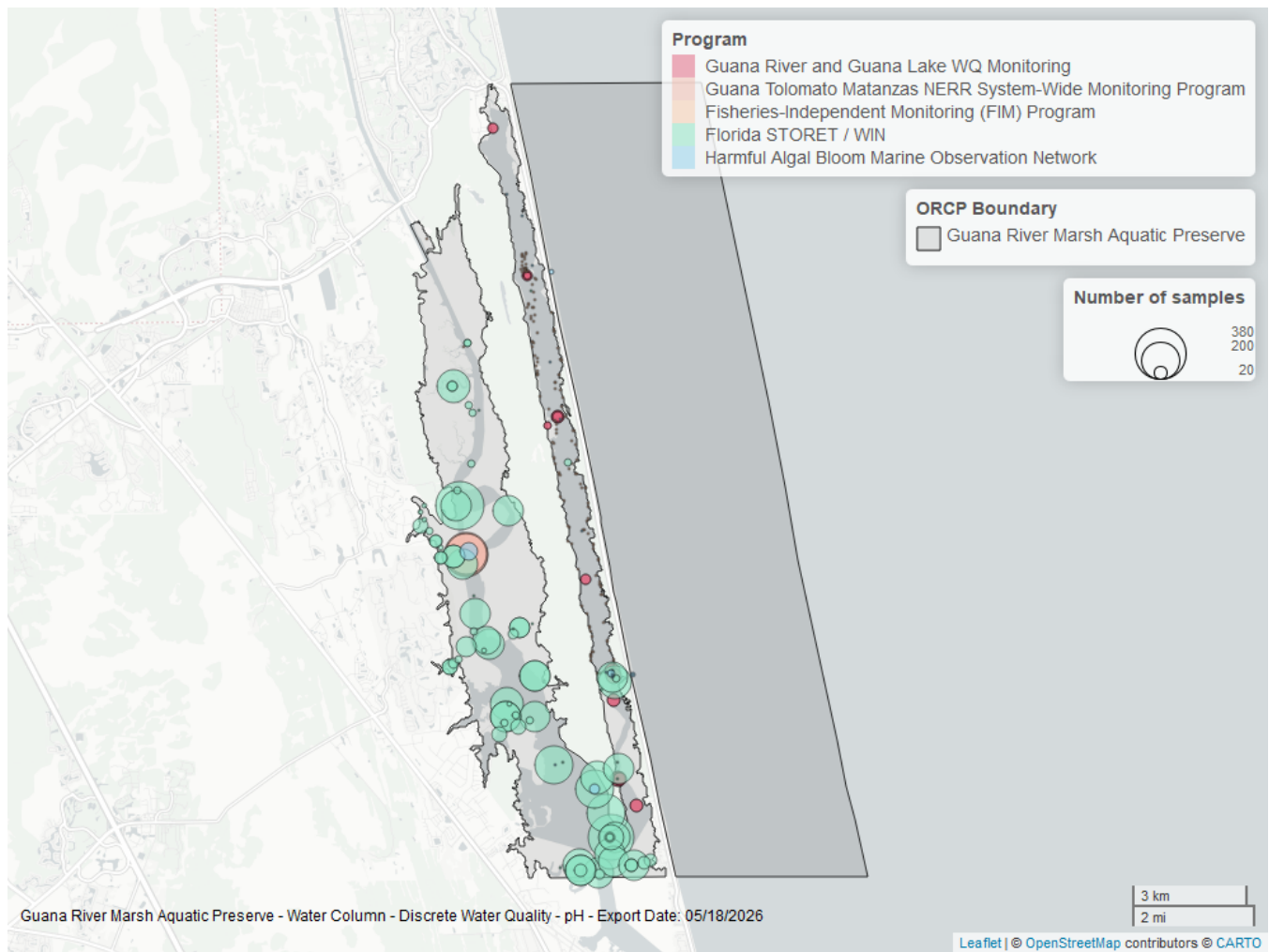


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for pH

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 5506   | 1995    | 2025    |
| 4054      | 535    | 2002    | 2025    |
| 5014      | 299    | 2017    | 2024    |
| 69        | 152    | 2001    | 2010    |
| 95        | 99     | 2007    | 2018    |

**Program names:**

5002 - Florida STORET / WIN<sup>3</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

69 - Fisheries-Independent Monitoring (FIM) Program<sup>5</sup>

95 - Harmful Algal Bloom Marine Observation Network<sup>6</sup>

# Salinity - Discrete

## Seasonal Kendall-Tau Trend Analysis

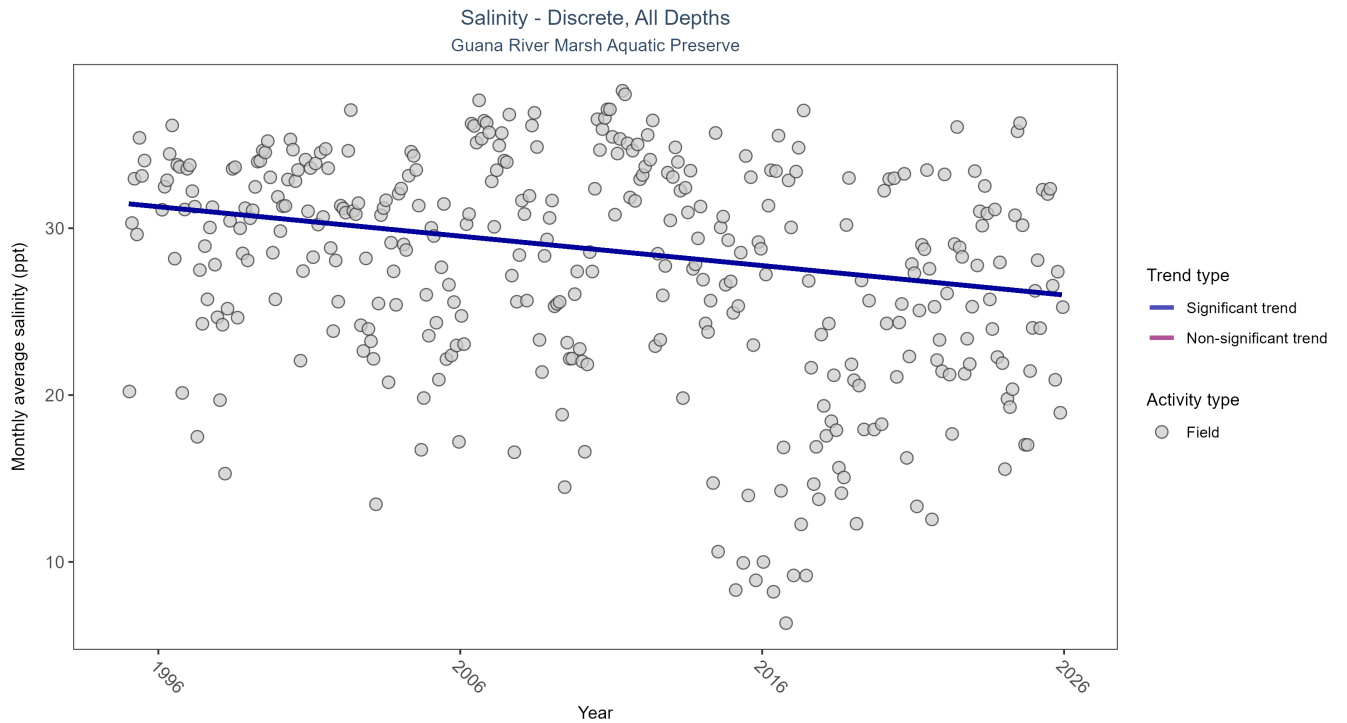


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Salinity

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|---------|---------------|-----------|---|
| All           | Decreasing trend  | 8491           | 31                  | 1995 - 2025      | 30.9                | -0.1902 | 31.4646       | -0.1764   | 0 |

Monthly average salinity decreased by 0.18 ppt per year.

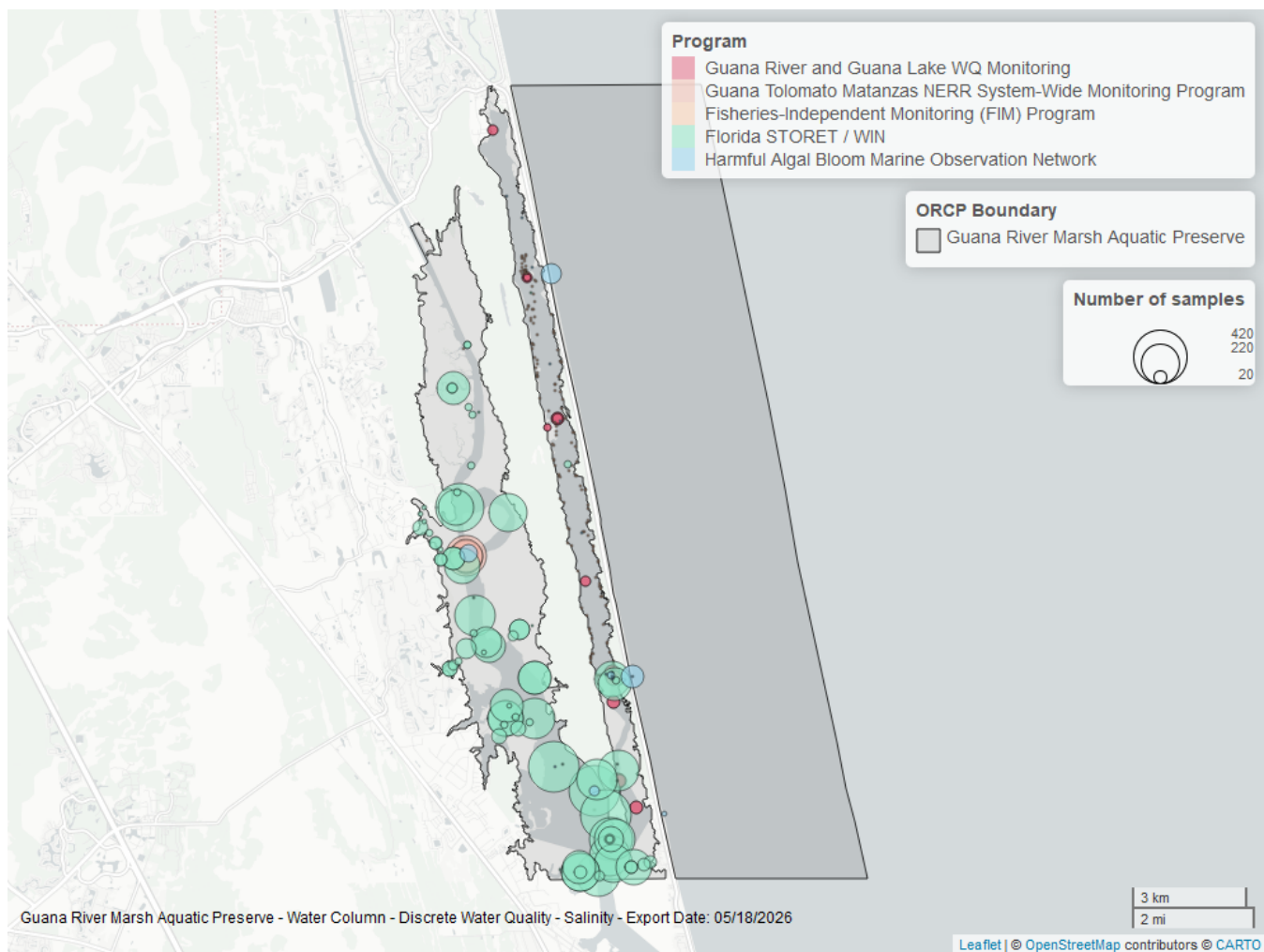


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Salinity

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 7399   | 1995    | 2025    |
| 4054      | 431    | 2002    | 2025    |
| 5014      | 299    | 2017    | 2024    |
| 95        | 236    | 1999    | 2018    |
| 69        | 152    | 2001    | 2010    |

**Program names:**

5002 - Florida STORET / WIN<sup>3</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

95 - Harmful Algal Bloom Marine Observation Network<sup>6</sup>

69 - Fisheries-Independent Monitoring (FIM) Program<sup>5</sup>

## Total Nitrogen - Discrete

### Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1)  $TN = TKN + NO3O2$ ;
- 2)  $TN = TKN + NO3 + NO2$ ;
- 3)  $TN = ORGN + NH4 + NO3O2$ ;
- 4)  $TN = ORGN + NH4 + NO2 + NO3$ ;
- 5)  $TN = TKN + NO3$ ;
- 6)  $TN = ORGN + NH4 + NO3$ ;

### Additional Information:

- Rules for use of sample fraction:
  - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
  - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
  - ParameterName = “Total Nitrogen”
  - SEACAR\_QAQCFlagCode = “1Q”
  - SEACAR\_QAQC\_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

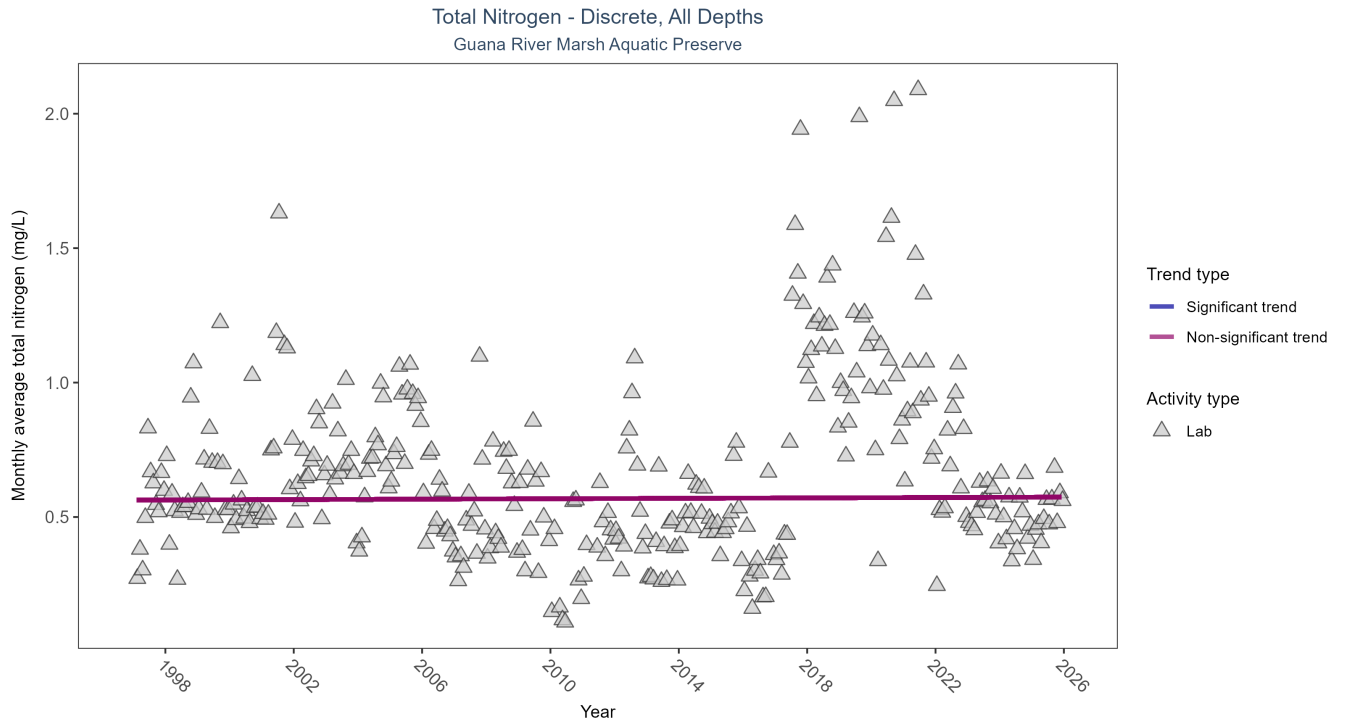


Figure 15: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

| Activity Type | Statistical Trend   | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|---------------------|----------------|---------------------|------------------|---------------------|--------|---------------|-----------|--------|
| Lab           | No detectable trend | 2055           | 29                  | 1997 - 2025      | 0.609               | 0.0115 | 0.563         | 0.0004    | 0.7571 |

Total nitrogen showed no detectable trend between 1997 and 2025.



Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Total Nitrogen

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 1175   | 1997    | 2025    |
| 5014      | 477    | 2017    | 2022    |
| 4054      | 428    | 2002    | 2025    |

#### Program names:

5002 - Florida STORET / WIN<sup>3</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

#### Total Phosphorus - Discrete

#### Seasonal Kendall-Tau Trend Analysis

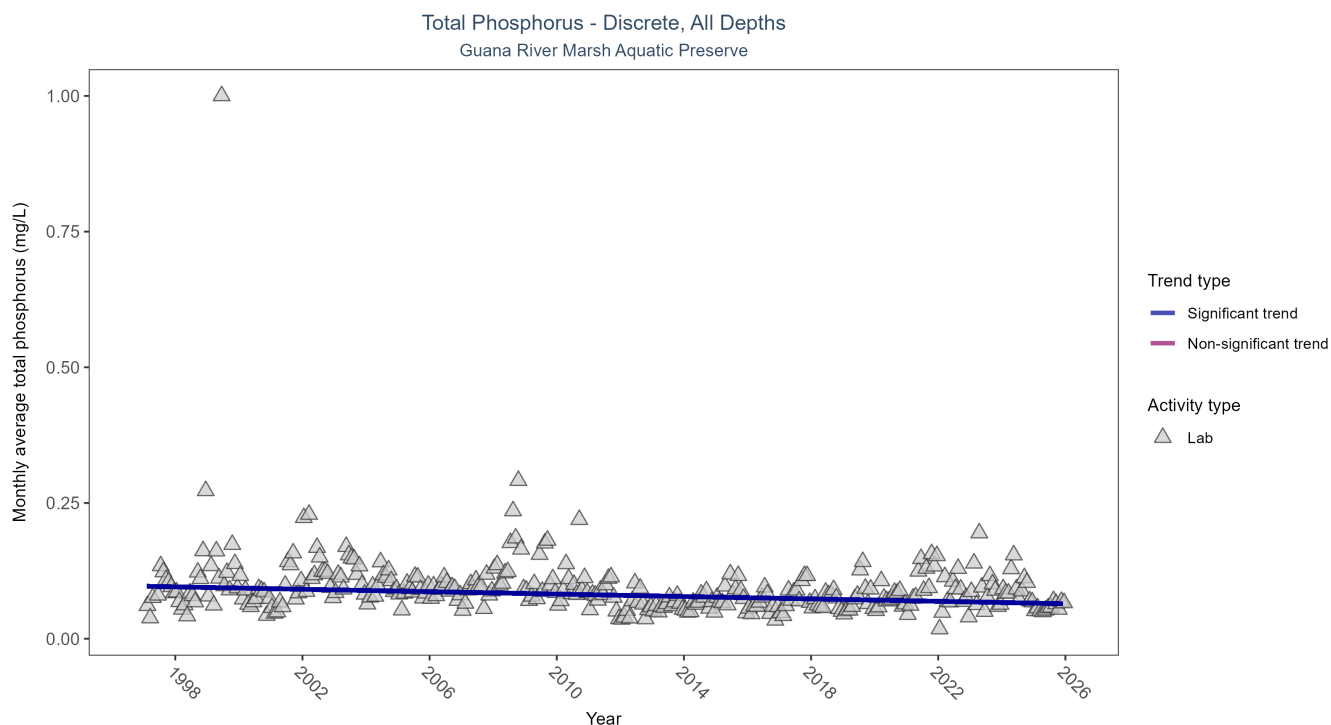


Figure 17: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|---------|---------------|-----------|---|
| Lab           | Decreasing trend  | 2974           | 29                  | 1997 - 2025      | 0.075               | -0.2302 | 0.0966        | -0.0011   | 0 |

Monthly average total phosphorus decreased by less than 0.01 mg/L per year.



Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Total Phosphorus

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 1799   | 1997    | 2025    |
| 5014      | 631    | 2017    | 2024    |
| 4054      | 608    | 2002    | 2025    |

#### Program names:

5002 - Florida STORET / WIN<sup>3</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

#### Turbidity - Discrete

#### Seasonal Kendall-Tau Trend Analysis

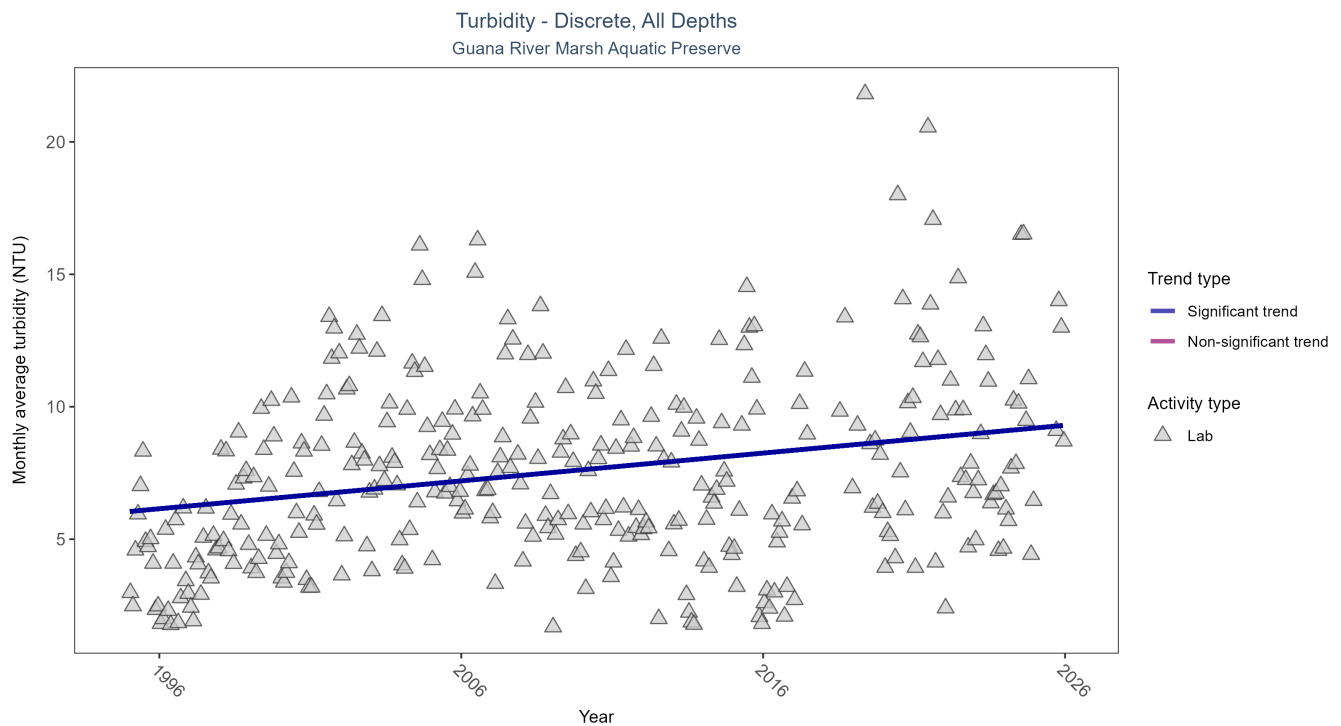


Figure 19: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 24: Seasonal Kendall-Tau Trend Analysis for Turbidity

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|--------|---------------|-----------|---|
| Lab           | Increasing trend  | 5030           | 31                  | 1995 - 2025      | 5.2                 | 0.2263 | 6.0441        | 0.1051    | 0 |

Monthly average turbidity increased by 0.11 NTU per year, indicating a decrease in water clarity.

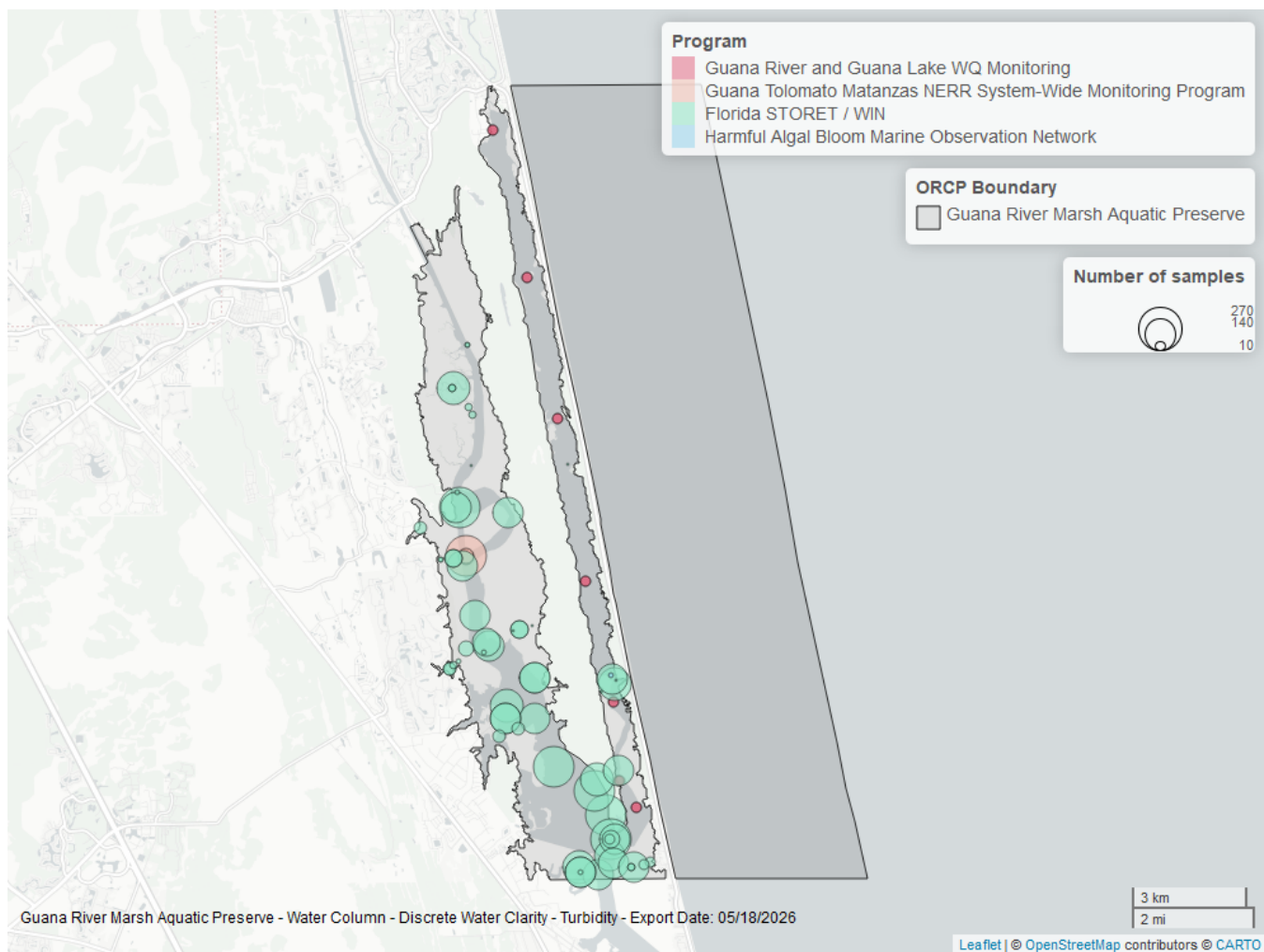


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 25: Programs contributing data for Turbidity

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 4901   | 1995    | 2025    |
| 4054      | 288    | 2002    | 2021    |
| 5014      | 126    | 2018    | 2022    |
| 95        | 4      | 2012    | 2012    |

#### Program names:

5002 - Florida STORET / WIN<sup>3</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

95 - Harmful Algal Bloom Marine Observation Network<sup>6</sup>

#### Water Temperature - Discrete

#### Seasonal Kendall-Tau Trend Analysis

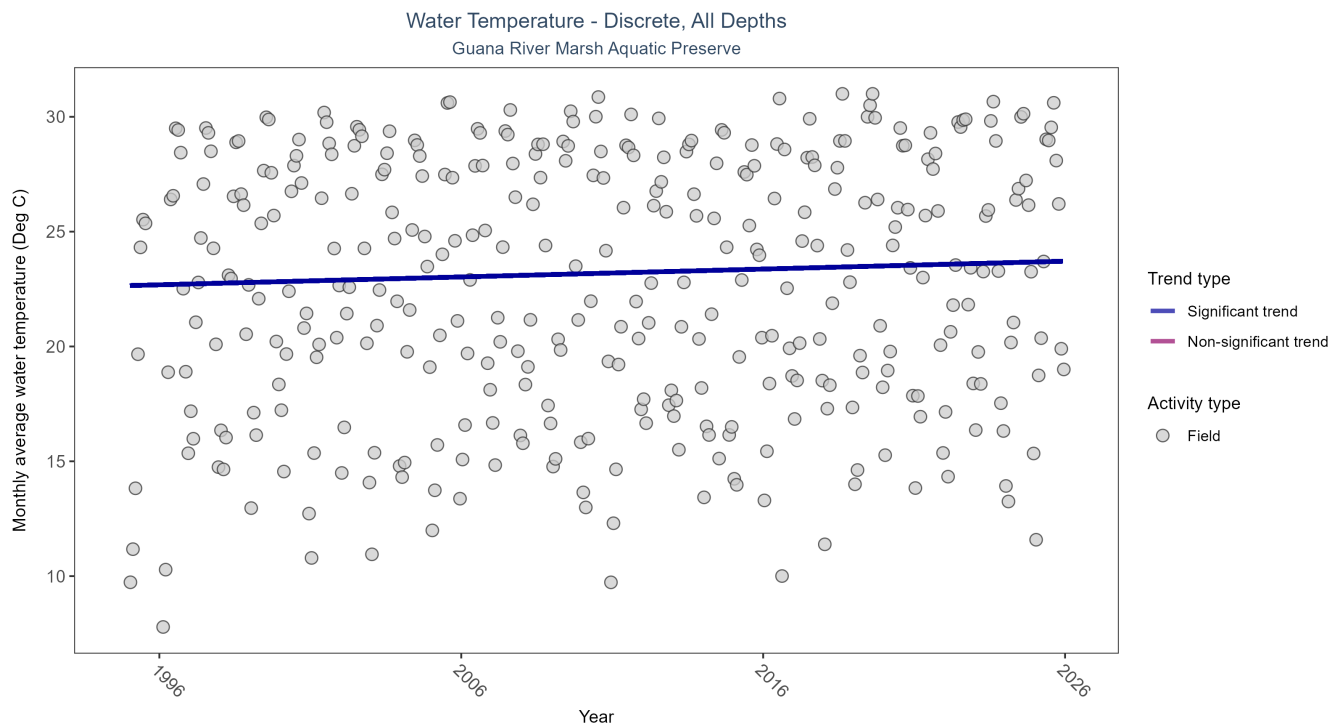


Figure 21: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 26: Seasonal Kendall-Tau Trend Analysis for Water Temperature

| Activity Type | Statistical Trend | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|-------------------|----------------|---------------------|------------------|---------------------|--------|---------------|-----------|--------|
| Field         | Increasing trend  | 8398           | 31                  | 1995 - 2025      | 23.1                | 0.1337 | 22.6504       | 0.0343    | 0.0003 |

Monthly average water temperature increased by  $0.03^{\circ}\text{C}$  per year.

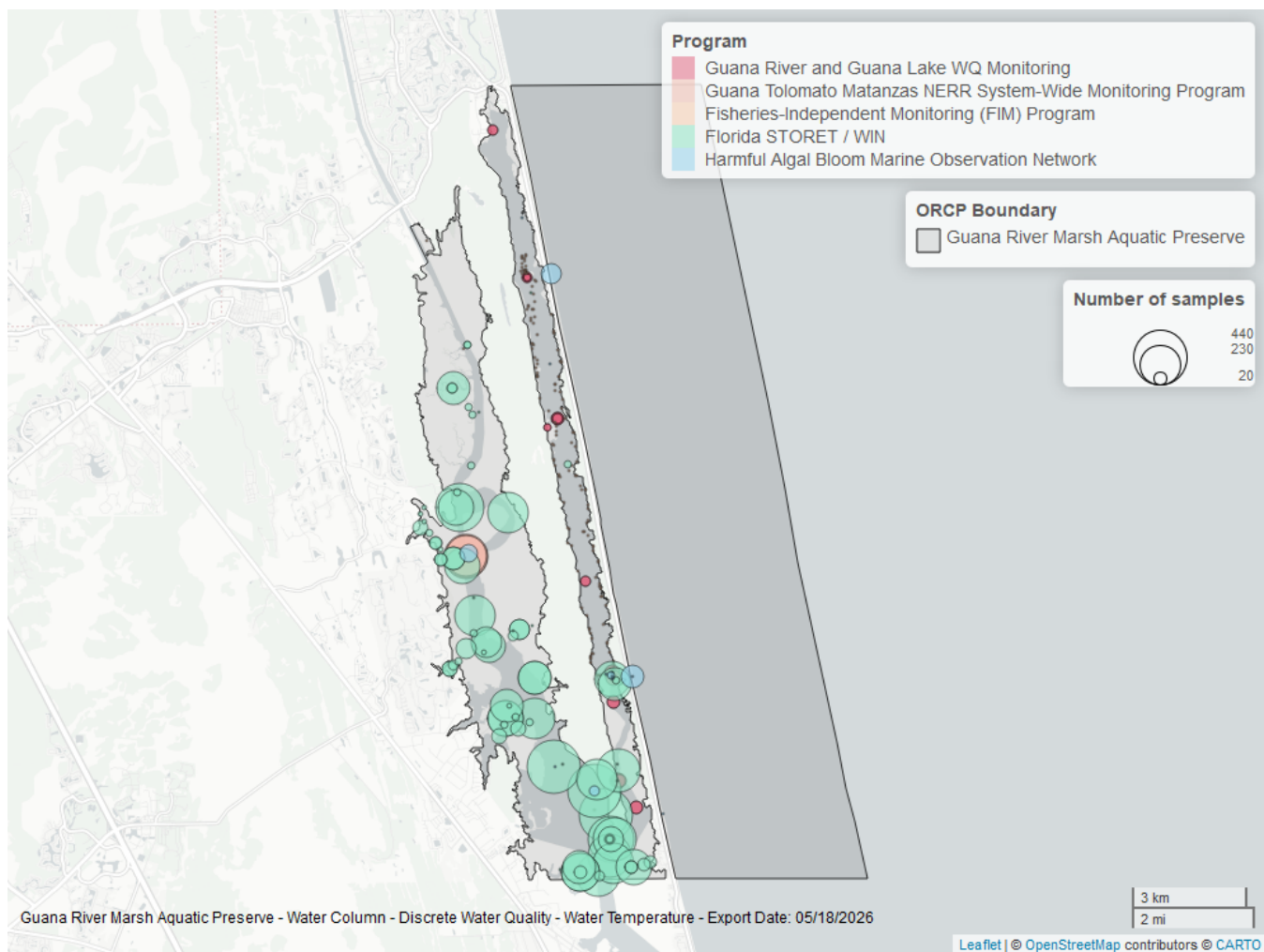


Figure 22: Map showing location of discrete water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 27: Programs contributing data for Water Temperature

| ProgramID | N_Data | YearMin | YearMax |
|-----------|--------|---------|---------|
| 5002      | 7609   | 1995    | 2025    |
| 4054      | 545    | 2002    | 2025    |
| 5014      | 299    | 2017    | 2024    |
| 95        | 228    | 2007    | 2018    |
| 69        | 152    | 2001    | 2010    |

**Program names:**

5002 - Florida STORET / WIN<sup>3</sup>

4054 - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>

5014 - Guana River and Guana Lake Water Quality Monitoring<sup>2</sup>

95 - Harmful Algal Bloom Marine Observation Network<sup>6</sup>

69 - Fisheries-Independent Monitoring (FIM) Program<sup>5</sup>

## Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined\_WQ\_WC\_NUT\_cont\_Chlorophyll\_a\_Uncorrected\_for\_Pheophytin-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Dissolved\_Oxygen-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Dissolved\_Oxygen\_Saturation-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Fluorescent\_Dissolved\_Organic\_Matter-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_pH-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Salinity-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Specific\_Conductivity-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Turbidity-2026-Mar-06.txt*
- *Combined\_WQ\_WC\_NUT\_cont\_Water\_Temperature-2026-Mar-06.txt*

### Continuous monitoring locations in Guana River Marsh Aquatic Preserve

Table 28: Station overview for Continuous parameters by Program

| <i>ProgramID</i> | <i>ProgramLocationID</i> | <i>Years of Data</i> | <i>Use in Analysis</i> | <i>Parameters</i>                                                                                                     |
|------------------|--------------------------|----------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------|
| 4054             | gtmpiwq                  | 26                   | TRUE                   | pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity |
| 5062             | 872-0494                 | 2                    | FALSE                  | Water Temperature , Salinity , Specific Conductivity                                                                  |

#### Program names:

[4054](#) - Guana Tolomato Matanzas National Estuarine Research Reserve System-Wide Monitoring Program<sup>1</sup>  
[5062](#) - FDEP Bureau of Survey and Mapping Continuous Water Quality Program<sup>7</sup>

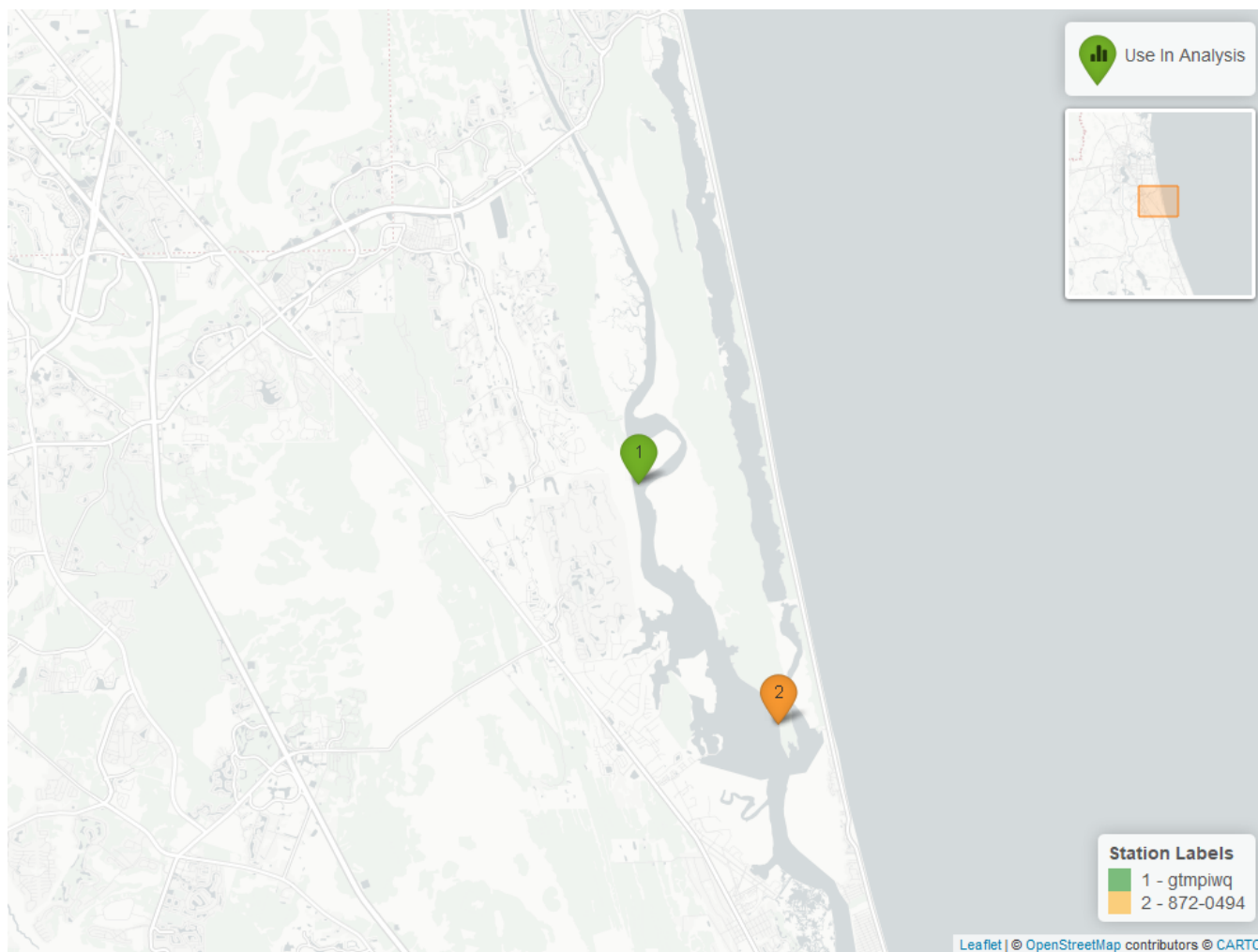


Figure 23: Map showing continuous water quality sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

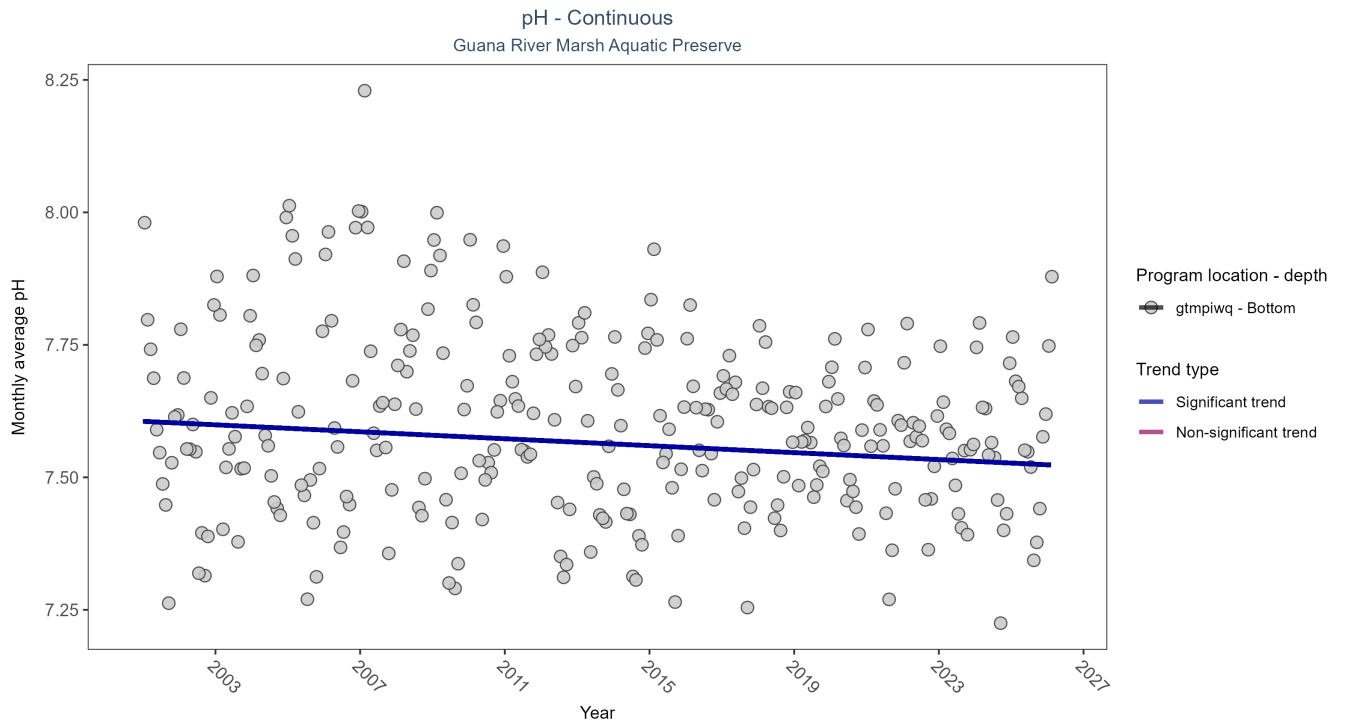


Figure 24: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for pH - All Stations

| Program Location | Statistical Trend              | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|---|
| gtmpiwq          | Significantly decreasing trend | 696449         | 26                  | 2001 - 2026      | 7.6                 | -0.17 | 7.61          | 0         | 0 |

At one program location, monthly average pH decreased by less than 0.01 pH units per year.

Dissolved Oxygen Saturation - Continuous

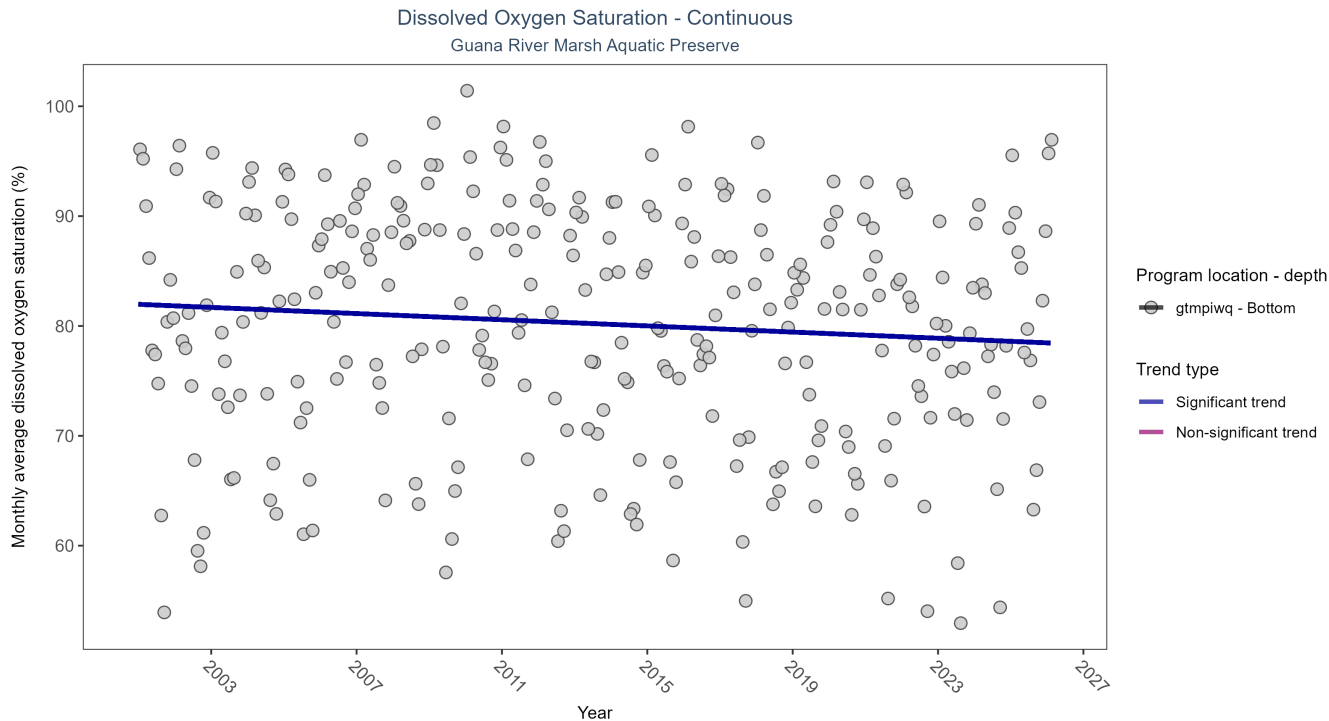


Figure 25: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

| Program Location | Statistical Trend              | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|---|
| gtmpiwq          | Significantly decreasing trend | 705219         | 26                  | 2001 - 2026      | 82.2                | -0.15 | 81.97         | -0.14     | 0 |

At one program location, monthly average dissolved oxygen saturation decreased by 0.14% per year.

Dissolved Oxygen - Continuous

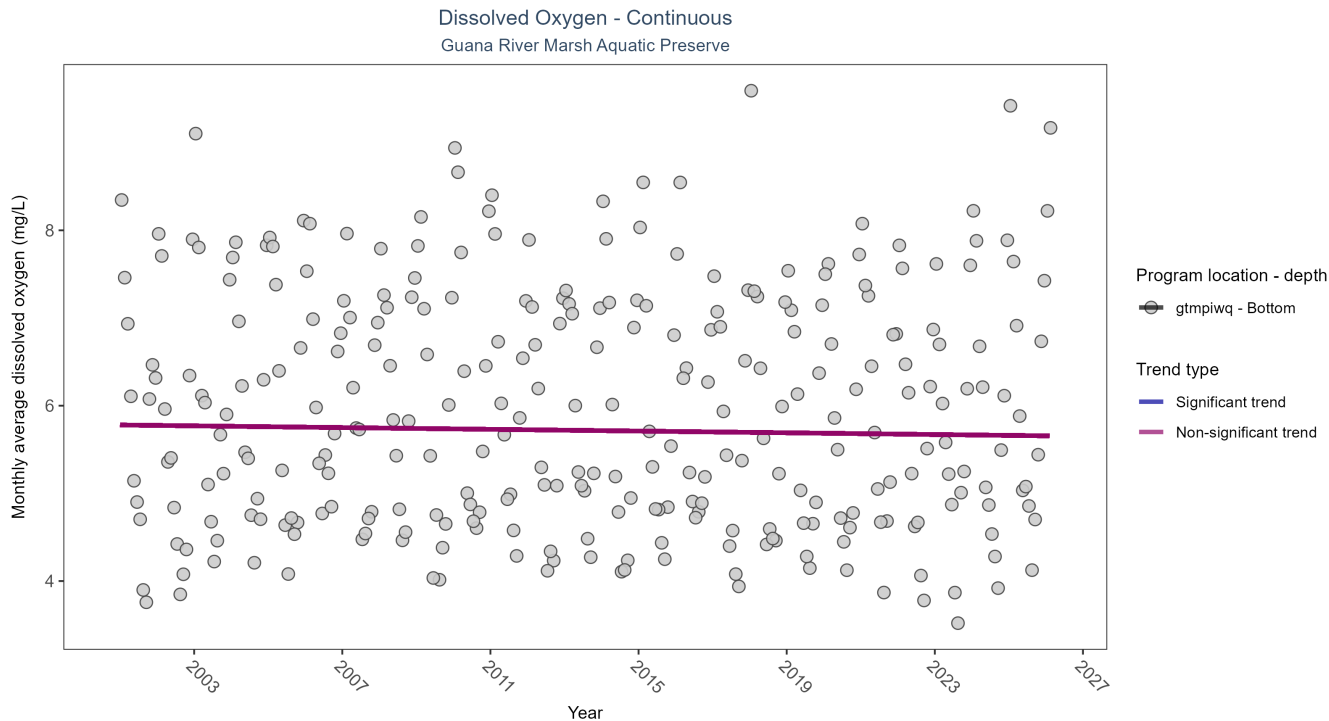


Figure 26: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

| Program Location | Statistical Trend    | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P    |
|------------------|----------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|------|
| gtmpiwq          | No significant trend | 699401         | 26                  | 2001 - 2026      | 5.9                 | -0.06 | 5.78          | 0         | 0.13 |

No detectable change in monthly average dissolved oxygen was observed at one location.

Turbidity - Continuous

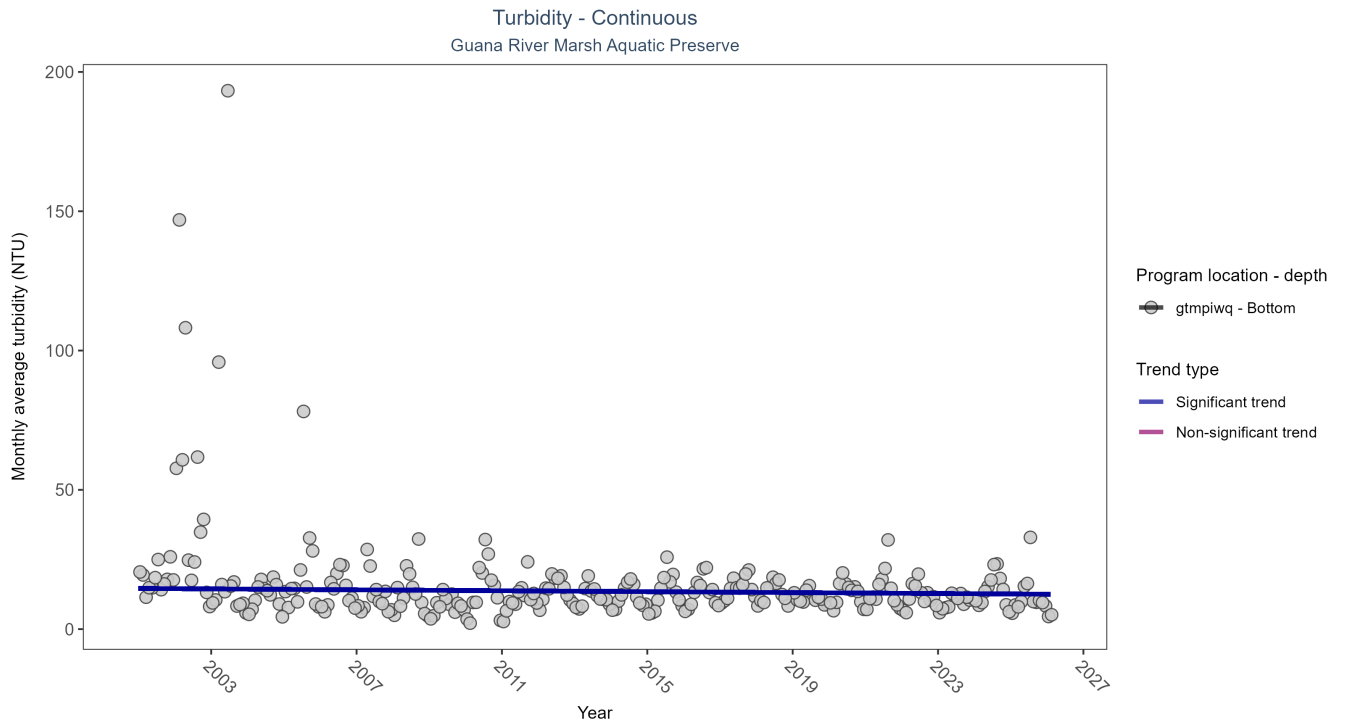


Figure 27: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Turbidity - All Stations

| Program Location | Statistical Trend              | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|---|
| gtmpiwq          | Significantly decreasing trend | 679520         | 26                  | 2001 - 2026      | 10                  | -0.12 | 14.61         | -0.08     | 0 |

At one program location, monthly average turbidity decreased by 0.08 NTU per year.

Water Temperature - Continuous

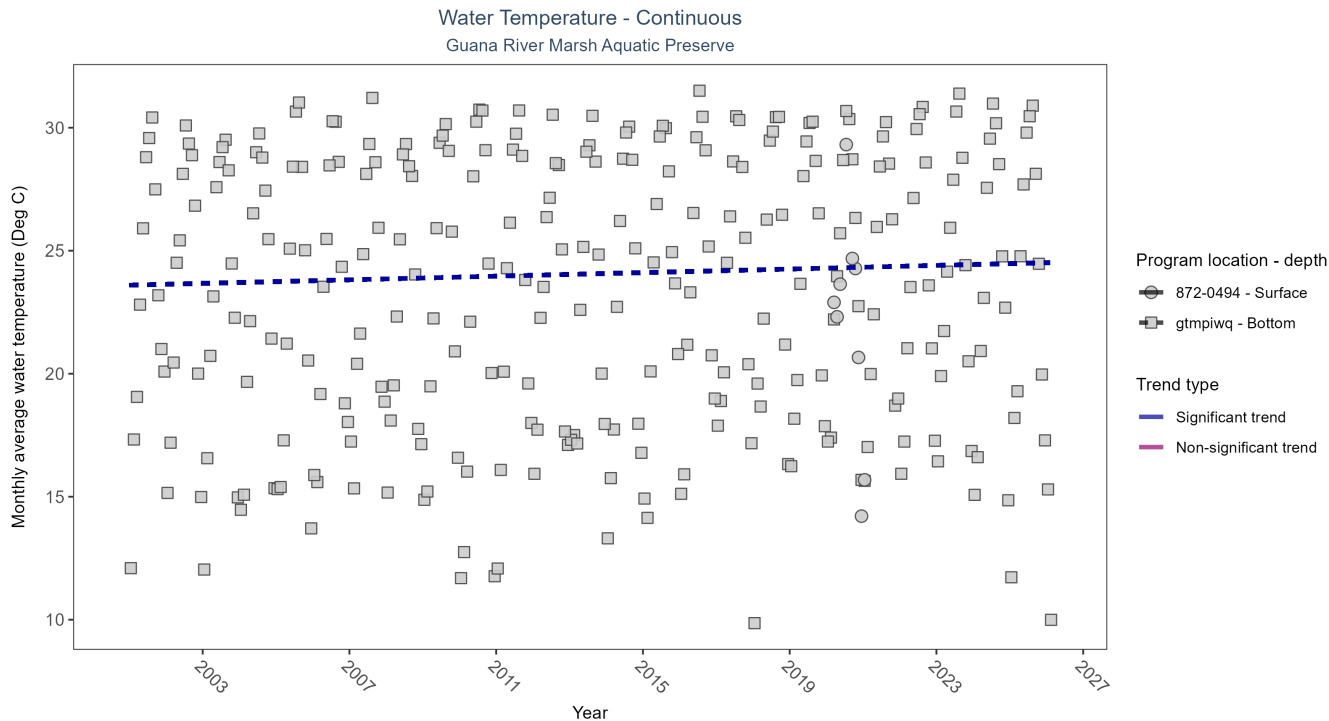


Figure 28: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Water Temperature - All Stations

| Program Location | Statistical Trend                    | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------------|----------------|---------------------|------------------|---------------------|------|---------------|-----------|---|
| gtmpiwiq         | Significantly increasing trend       | 753811         | 26                  | 2001 - 2026      | 24.20               | 0.18 | 23.6          | 0.04      | 0 |
| 872-0494         | Insufficient data to calculate trend | 35473          | 2                   | 2020 - 2021      | 22.34               | -    | -             | -         | - |

At one program location, monthly average water temperature increased by 0.04°C per year. There was insufficient data to fit a model for one location.

Salinity - Continuous

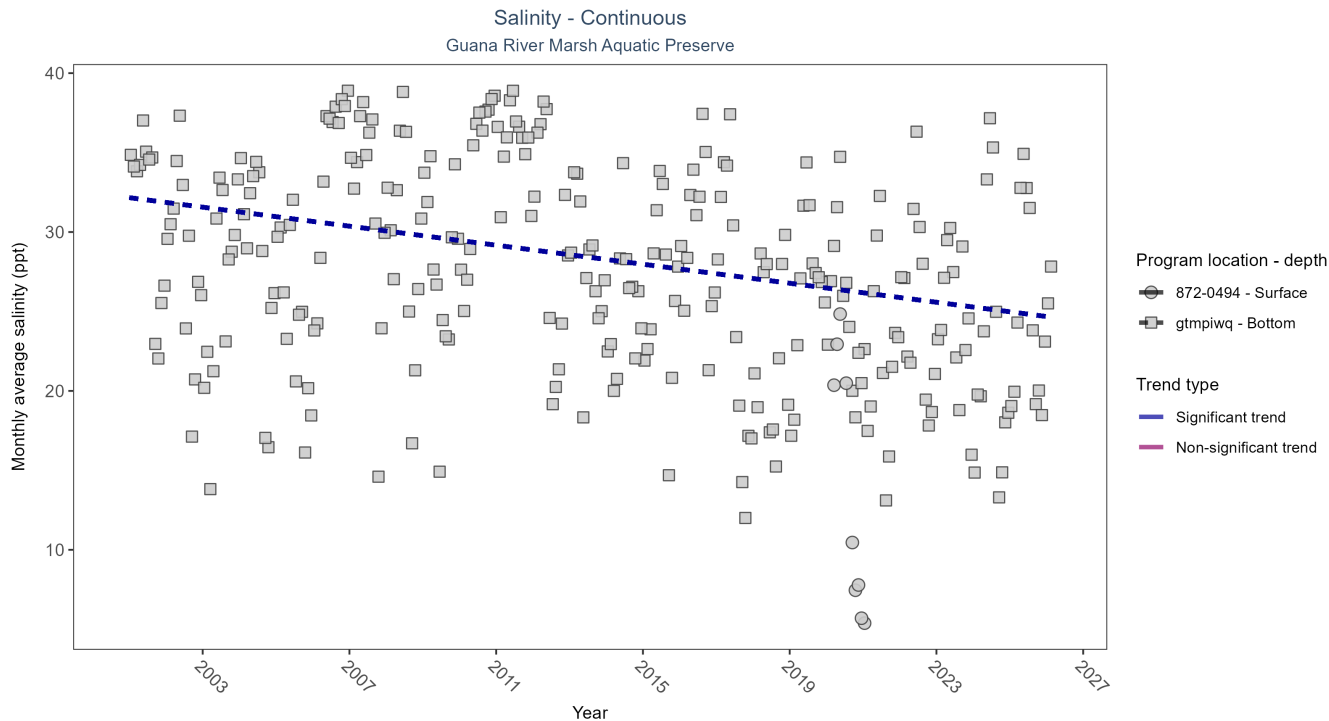


Figure 29: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 34: Seasonal Kendall-Tau Results for Salinity - All Stations

| Program Location | Statistical Trend                    | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|---|
| gtmpiwq          | Significantly decreasing trend       | 703518         | 26                  | 2001 - 2026      | 27.70               | -0.26 | 32.16         | -0.3      | 0 |
| 872-0494         | Insufficient data to calculate trend | 34918          | 2                   | 2020 - 2021      | 8.99                | -     | -             | -         | - |

At one program location, monthly average salinity decreased by 0.30 ppt per year. There was insufficient data to fit a model for one location.

Specific Conductivity - Continuous

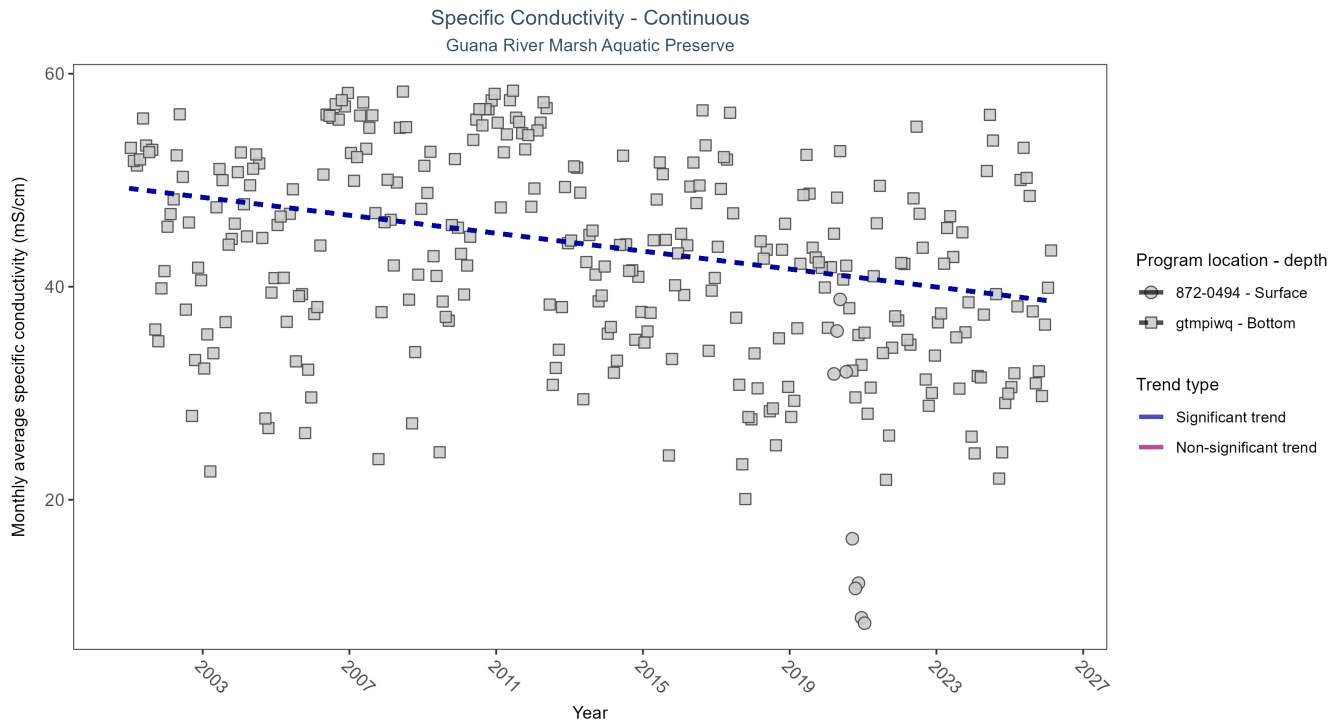


Figure 30: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 35: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

| Program Location | Statistical Trend                    | No. of Samples | No. Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P |
|------------------|--------------------------------------|----------------|---------------------|------------------|---------------------|-------|---------------|-----------|---|
| gtmpiwq          | Significantly decreasing trend       | 703518         | 26                  | 2001 - 2026      | 43.06               | -0.27 | 49.24         | -0.42     | 0 |
| 872-0494         | Insufficient data to calculate trend | 34894          | 2                   | 2020 - 2021      | 14.10               | -     | -             | -         | - |

At one program location, monthly average specific conductivity decreased by 0.42 mS/cm per year. There was insufficient data to fit a model for one location.

# Coastal Wetlands

The data file used is: All\_CW\_Parameters-2026-Jun-04.txt

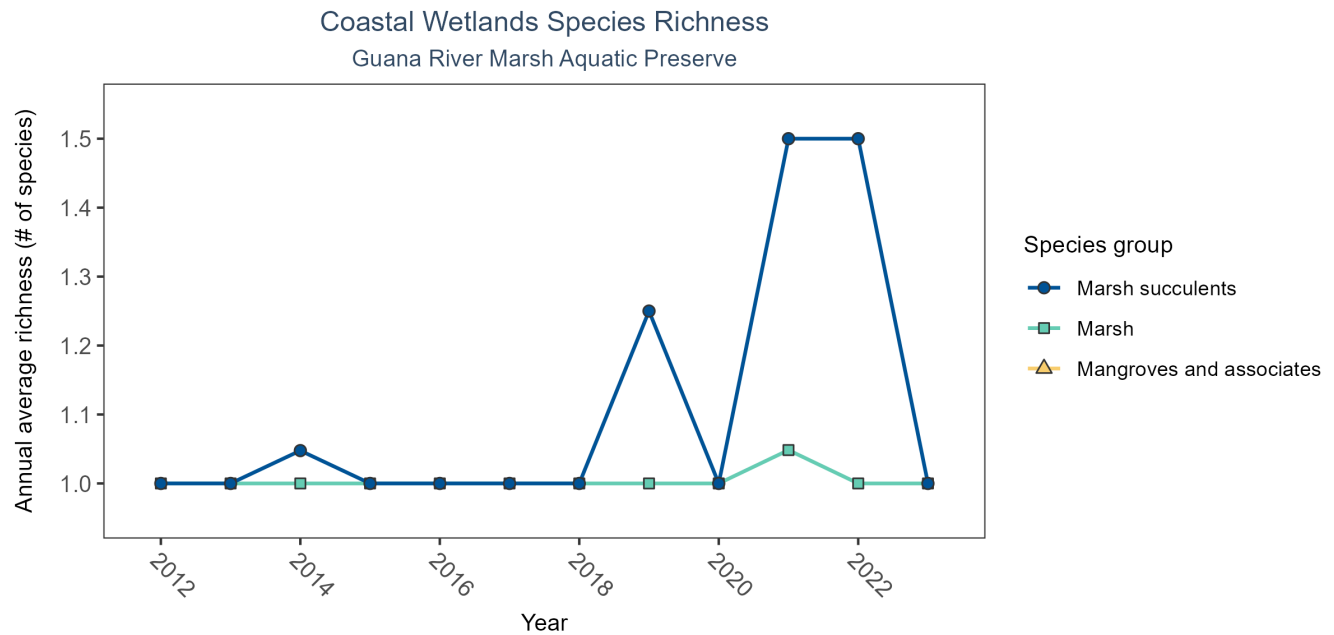


Figure 31: Line graph of annual average coastal wetlands species richness over time for mangroves and associates (triangles), marsh (squares), and marsh succulents (circles). If the time series by species group included more than one year of observations, a line connects data points for visualization.

Table 36: Coastal Wetlands Species Richness

| <i>Species Group</i> | <i>No. of Samples</i> | <i>No. Years with Data</i> | <i>Period of Record</i> | <i>Median No. of Taxa</i> | <i>Mean No. of Taxa</i> |
|----------------------|-----------------------|----------------------------|-------------------------|---------------------------|-------------------------|
| Marsh                | 795                   | 12                         | 2012 - 2023             | 1                         | 1.0                     |
| Marsh succulents     | 58                    | 12                         | 2012 - 2023             | 1                         | 1.1                     |

Between 2012 and 2023, the median annual number of species for marsh was 1 based on 795 observations. Between 2012 and 2023, the median annual number of species for marsh succulents was 1 based on 58 observations.

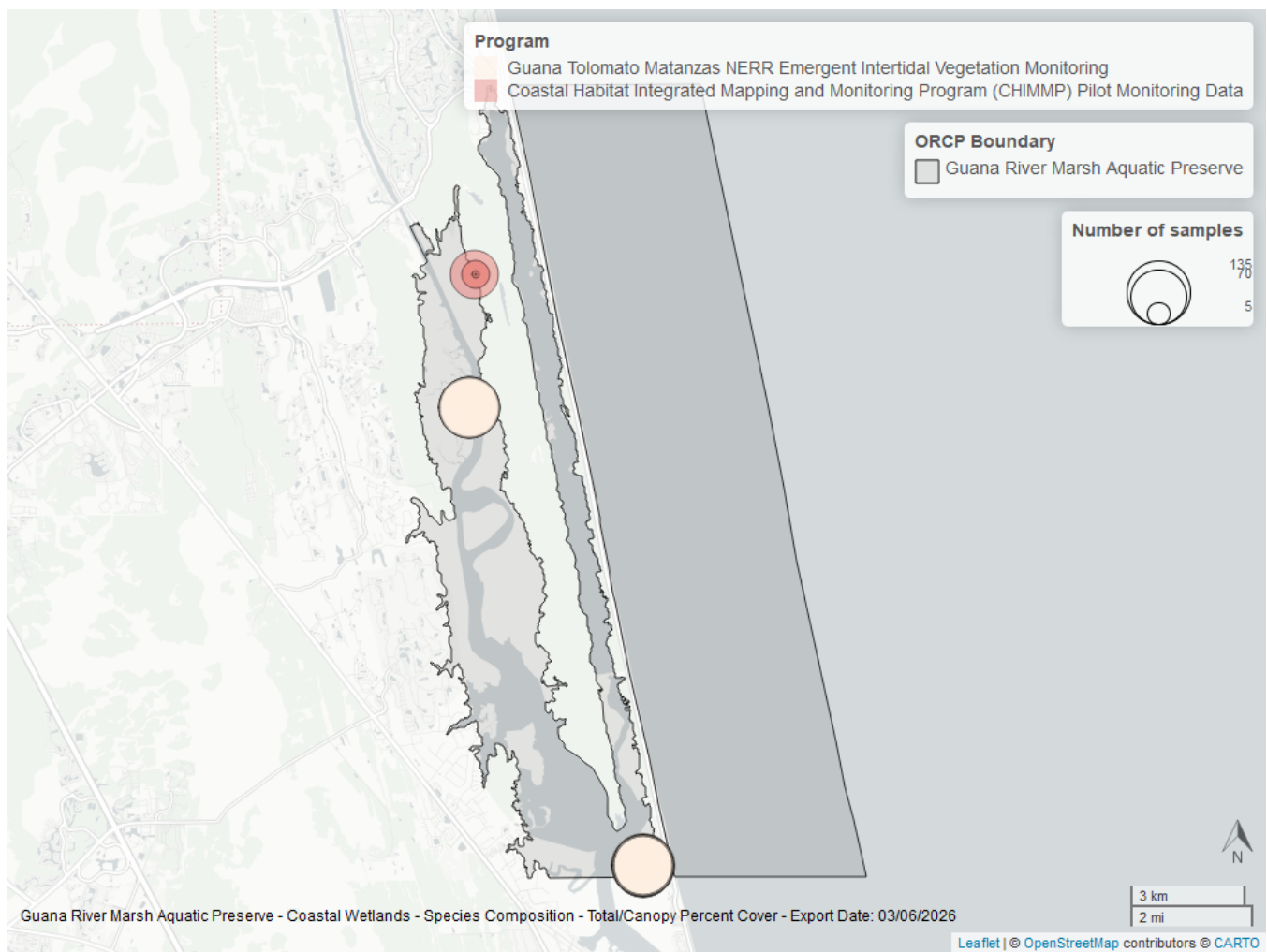


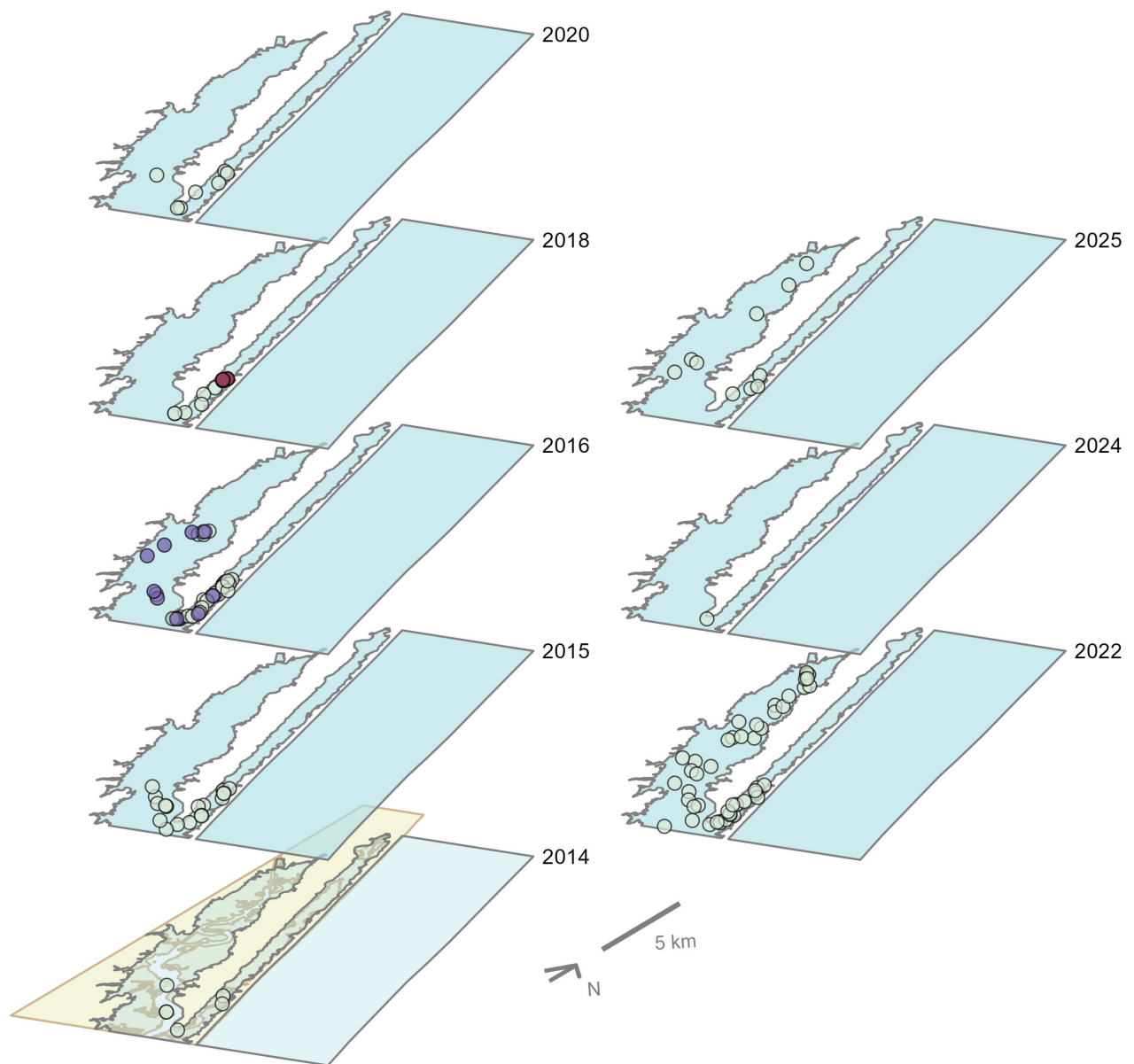
Figure 32: Map showing location of coastal wetlands sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Oyster

The data file used is: **All\_OYSTER\_Parameters-2026-Jun-04.txt**

## Guana River Marsh Aquatic Preserve

Oyster sample locations available in SEACAR by Program and Year



### Program name

- Guana Tolomato Matanzas NERR Oyster monitoring
- Northern Coastal Basins and Mosquito Lagoon Oyster Condition Assessment
- Historical Oyster Body Size (HOBS) Study

Figure 33: Maps showing the temporal scope of oyster sampling sites within the boundaries of *Guana River Marsh Aquatic Preserve* by program name.

Click [here](#) to view spatio-temporal plots on GitHub.

## Density

For natural reefs, density decreased by an average of 84.78 oysters per square meter per year.

### Natural

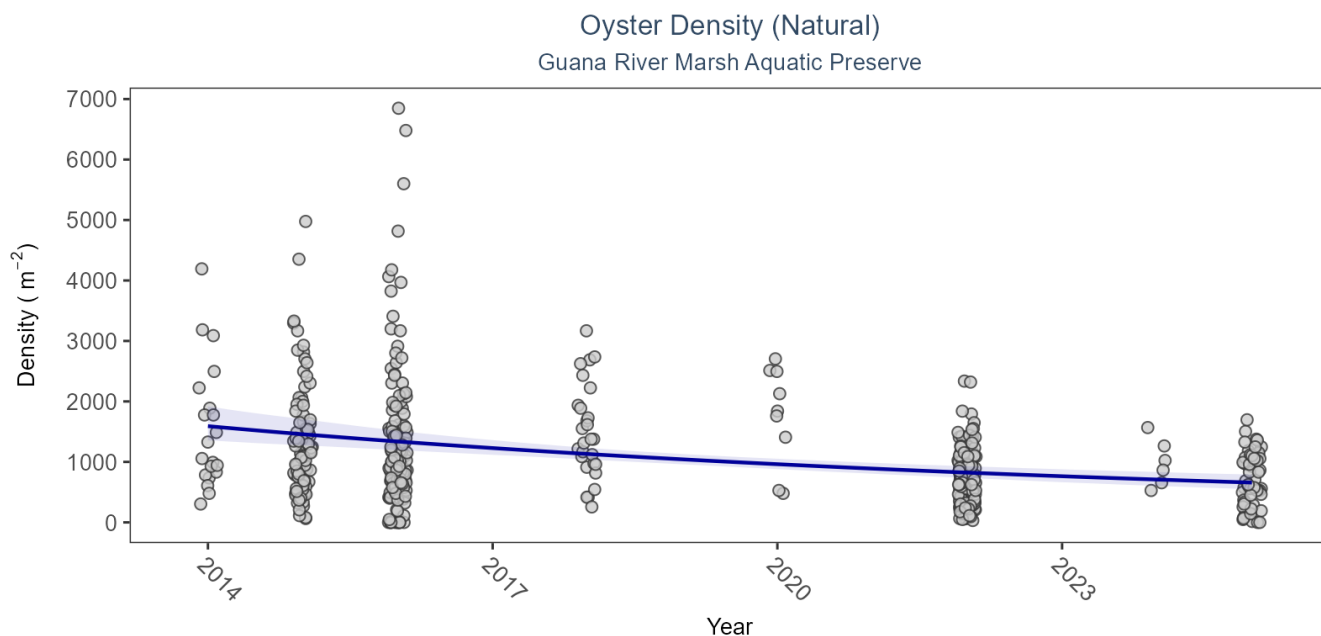


Figure 34: Scatter plot of oyster density over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Data points are jittered horizontally to reduce overlap.

Table 37: Model results for Oyster Density - Natural

| <i>Habitat Type</i> | <i>Shell Type</i> | <i>Statistical Trend</i> | <i>Period of Record</i> | <i>Estimate</i> | <i>Standard Error</i> | <i>Credible Interval</i> |
|---------------------|-------------------|--------------------------|-------------------------|-----------------|-----------------------|--------------------------|
| Natural             | Live Oysters      | Decreasing trend         | 2014 - 2025             | -84.78          | 16.27                 | -120.24 to -55.7         |

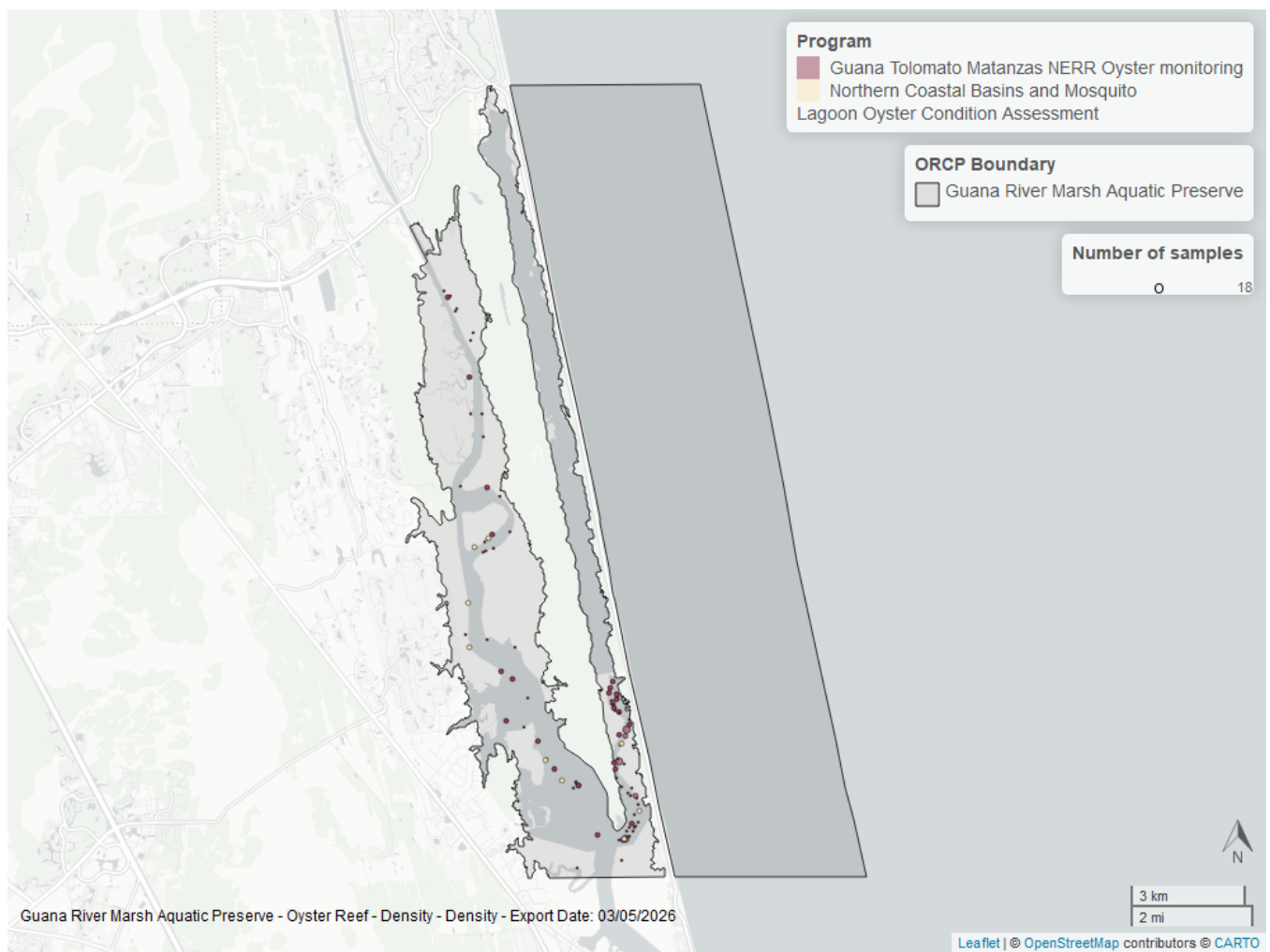


Figure 35: Map showing location of oyster density sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Percent Live

For natural reefs, percent live cover showed no detectable trend between 2014 and 2025.

Natural

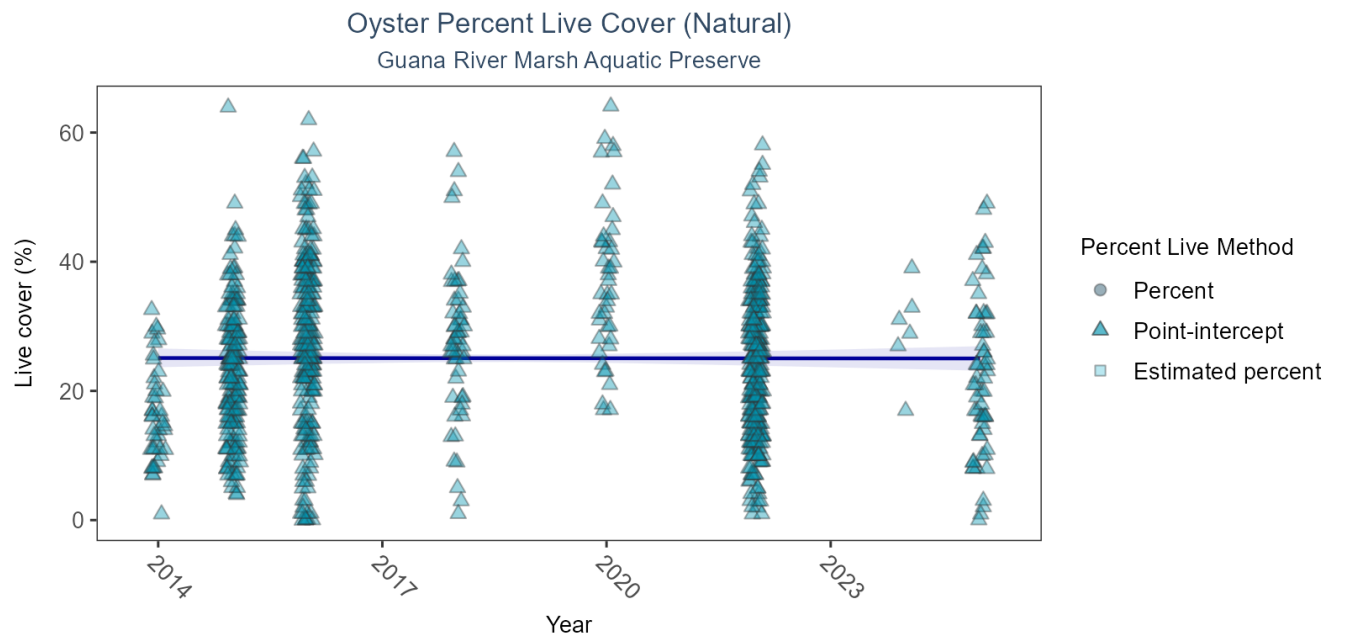


Figure 36: Scatter plot of percent live oysters over time. If the time series included five or more years with observations, an estimated trend (blue line) and a 95% credible interval (purple band) may also be plotted. Estimation method is represented as percent (circles), point-intercept (triangle), or estimated percent (square), with data points jittered horizontally to reduce overlap.

Table 38: Model results for Oyster Percent Live - Natural

| <i>Habitat Type</i> | <i>Shell Type</i> | <i>Statistical Trend</i> | <i>Period of Record</i> | <i>Estimate</i> | <i>Standard Error</i> | <i>Credible Interval</i> |
|---------------------|-------------------|--------------------------|-------------------------|-----------------|-----------------------|--------------------------|
| Natural             | Live Oysters      | No detectable trend      | 2014 - 2025             | -0.01           | 0.14                  | -0.29 to 0.28            |

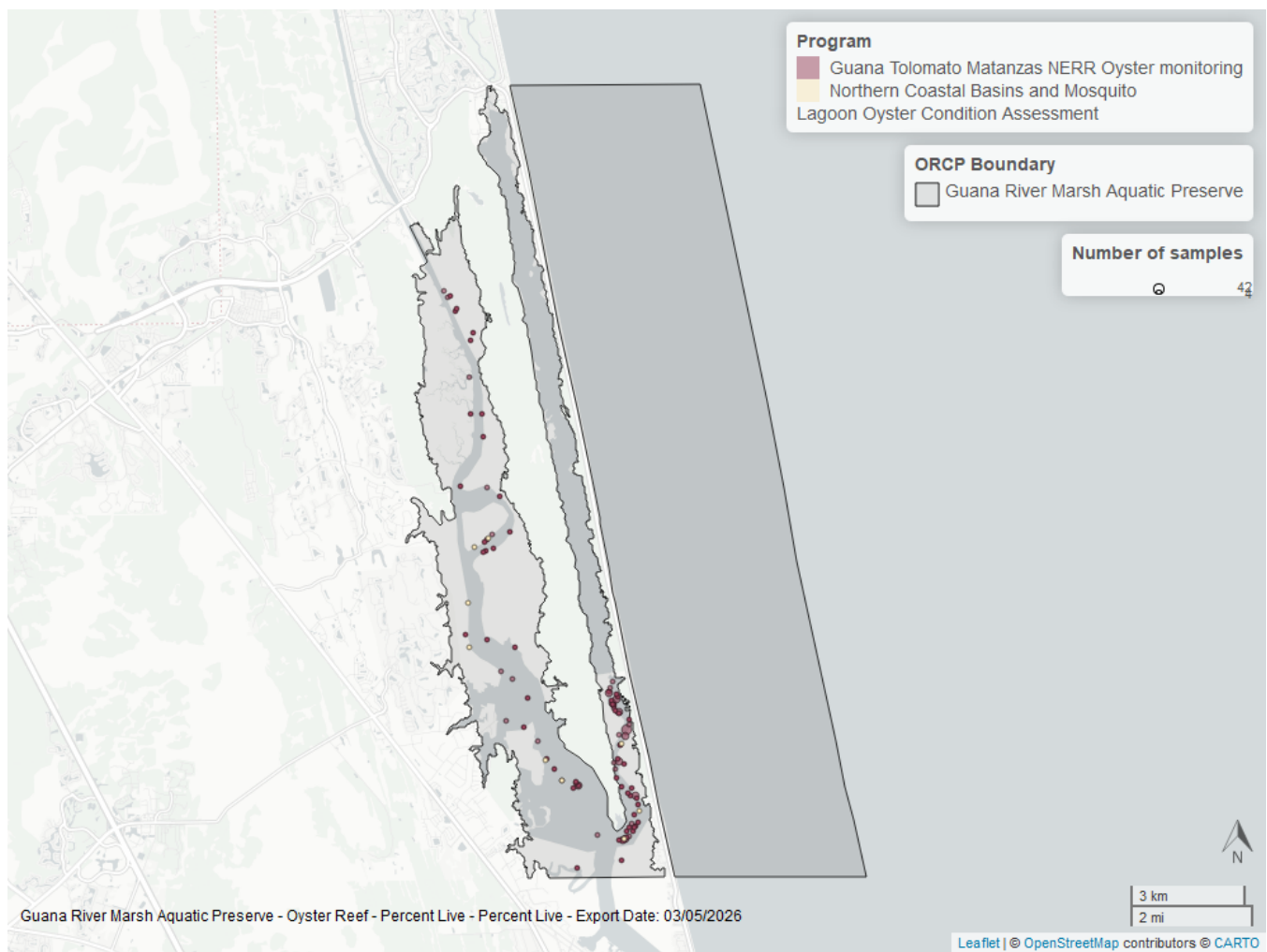


Figure 37: Map showing location of oyster percent live sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Shell Height

For natural reefs, annual average live oyster shell height in the 25-75mm size class increased by 0.1mm per year, and there was no detectable trend for live oysters in the  $\geq 75$ mm size class. Models are not run on dead oyster shell measurements.

## Natural

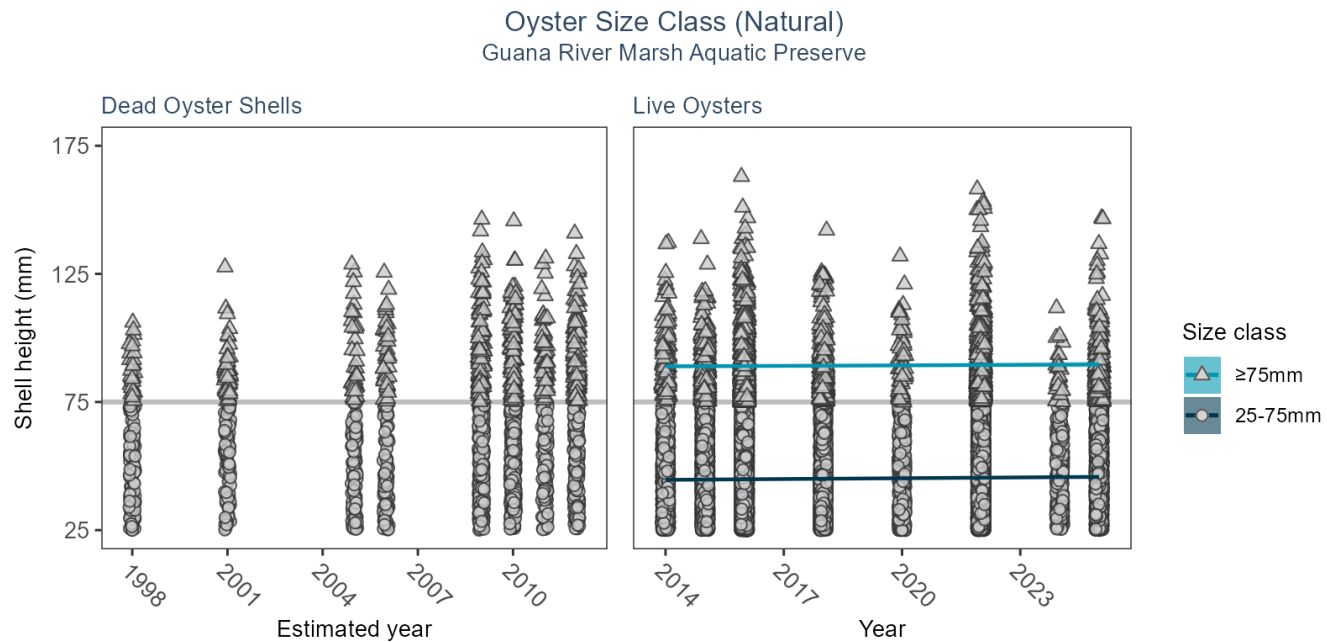


Table 39: Model results for Oyster Shell Height - Natural

| Habitat Type | Shell Type         | SizeClass      | Statistical Trend                  | Period of Record | No. of Samples | Estimate | Standard Error | Credible Interval |
|--------------|--------------------|----------------|------------------------------------|------------------|----------------|----------|----------------|-------------------|
| Natural      | Dead Oyster Shells | $>75\text{mm}$ | Model not run on dead oyster shell | 1998 - 2012      | 600            | -        | -              | -                 |
| Natural      | Dead Oyster Shells | 25-75mm        | Model not run on dead oyster shell | 1998 - 2012      | 1138           | -        | -              | -                 |
| Natural      | Live Oysters       | $>75\text{mm}$ | No detectable trend                | 2014 - 2025      | 2907           | 0.06     | 0.09           | -0.11 to 0.25     |
| Natural      | Live Oysters       | 25-75mm        | Increasing trend                   | 2014 - 2025      | 18613          | 0.10     | 0.05           | 0.01 to 0.2       |

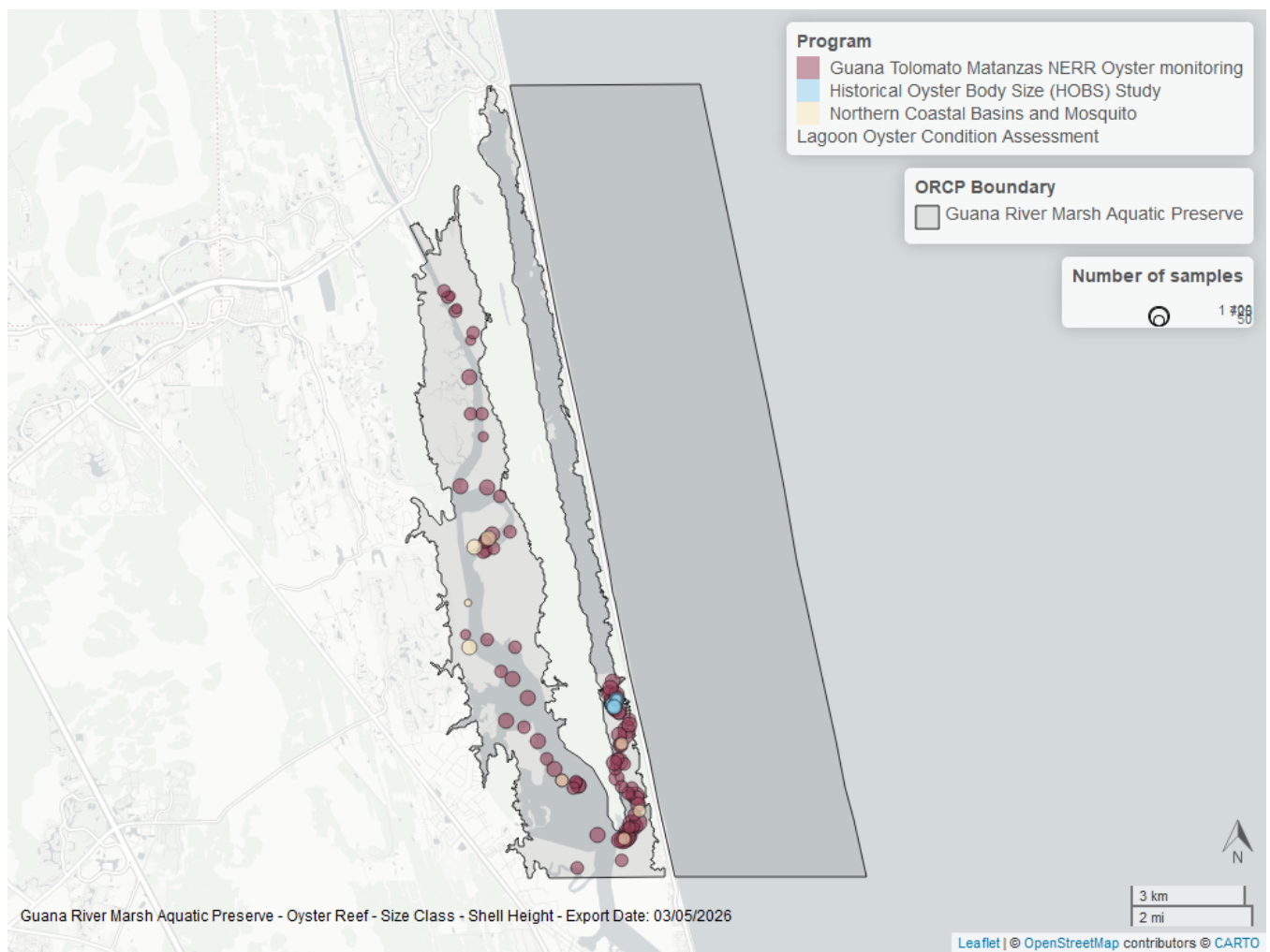


Figure 38: Map showing location of oyster shell height sampling locations within the boundaries of *Guana River Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Species list

|                                  |                                 |                                    |
|----------------------------------|---------------------------------|------------------------------------|
| Avicennia germinans <sup>1</sup> | Hypericum sp.                   | Salicornia ambigua <sup>1</sup>    |
| Bare substrate                   | Ilex vomitoria                  | Smilax bona-nox                    |
| Batis maritima <sup>1</sup>      | Iva frutescens                  | Spartina alterniflora <sup>1</sup> |
| Coleataenia anceps               | Juncus roemerianus <sup>1</sup> | Vitis sp.                          |
| Dichanthelium sp.                | Quercus virginiana              | Woody debris                       |
| Distichlis spicata <sup>1</sup>  | Sabal palmetto                  | Avicennia germinans <sup>1</sup>   |

1 - Coastal Wetlands

## References

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2. Florida Department of Environmental Protection (DEP); Office of Resilience and Coastal Protection (RCP); Guana Tolomato Matanzas National Estuarine Research Reserve (GTMNERR); Friends of GTM; Florida Fish and Wildlife Conservation Commission (FWC). [Guana River and Guana Lake Water Quality Monitoring](#) . (2024).
3. Florida Department of Environmental Protection (DEP). [Florida STORET / WIN](#). (2024).
4. U.S. Environmental Protection Agency (EPA); Office of Water; National Oceanic and Atmospheric Administration (NOAA); U.S. Geological Survey (USGS); U.S. Fish and Wildlife Service (USFWS); National Estuary Program (NEP); coastal states. [National Aquatic Resource Surveys, National Coastal Condition Assessment](#). (2021).
5. Florida Fish and Wildlife Conservation Commission (FWC). [Fisheries-Independent Monitoring \(FIM\) Program](#). (2022).
6. Florida Fish and Wildlife Conservation Commission (FWC); Florida Fish and Wildlife Research Institute (FWRI). [Harmful Algal Bloom Marine Observation Network](#). (2018).
7. Florida Department of Environmental Protection (DEP); Division of State Lands; Bureau of Survey and Mapping. [FDEP Bureau of Survey and Mapping Continuous Water Quality Program](#). (2021).